Healthy vaccinee and vaccination misclassification biases in observational health data mask Dutch real-world Covid-19 vaccine effectiveness and safety

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Abstract

Background: We analyze a large Dutch national observational mortality and Covid-19 vaccination cohort with the goals of (re)assessing real-world vaccine effectiveness (VE) and performing bias analysis. We compare against earlier work based on the same dataset that reported very high effectiveness and safety.

Methods: Primarily, we focus on the first Covid vaccination round (one or two doses) and its short-term and longer-term effects. As primary method we use standard epidemiological survival analysis combined with stratification and matching based on multiple available population and health indicators to correct for confounding. We also perform complementary, calendar time-based analyses using diverse population and health-related data, as well as epidemiological sanity checks focused on negative control data and on data integrity. Finally, we use quantitative bias analysis (QBA) techniques to better understand, quantify, and (approximately) adjust for identified systematic biases. In all analyses, we look at Covid, non-Covid, and (especially) all-cause mortality (ACM) effects.

Results and Discussion: Our results show unrealistically high mortality among individuals classified as unvaccinated during and after a vaccination round (starting immediately after dose 1). These effects cannot realistically be attributed to Covid-19 disease or prevention. The results and supporting evidence and analyses strongly suggest two large statistical artifacts: residual, frailty-related healthy vaccinee bias (despite matching) and vaccination status misclassification bias. These biases are conflated and together have a strong and largely disqualifying effect on the calculated mortality and VE results from our primary analysis, and likely also on previously reported results based on the same dataset. They are anomalies that are very hard to correct without additional fine-grained and high-integrity data. Two QBA breakdowns lend additional credence to these findings, but they can only approximately quantify and adjust for these biases.

Conclusions: Realistic age-stratified Covid-19 vaccine effectiveness and safety profiles in this dataset are masked by these large anomalies. Analysis of these anomalies suggests that they lead to overestimation of Covid vaccination real-world protection, which is likely considerably lower than previously described.

1. Introduction

Phase 3 trials of various vaccines designed to protect against Coronavirus disease (Covid-19) have demonstrated protection against symptomatic illness [1, 2, 3]. However, vaccines that protect against a target disease can still fail to show efficacy against all-cause mortality (ACM) [4, 5, 6, 7], have serious side effects in certain subgroups [8, 9], or even potentially lead to net negative effects [10]. Therefore, it is important to also evaluate vaccines against all-cause morbidity and mortality, or generally speaking, more general outcome metrics beyond the target disease. Unfortunately, few clinical trials are designed to study this extensively [7, 10].

During the Covid-19 pandemic there was an international effort to accelerate vaccine development, and conditional marketing authorization did not require that vaccines demonstrate positive effects against ACM [7, 11, 12]. Re-analysis of trial data with respect to ACM effects was inconclusive [6]. Conceivably ACM-based endpoints could have been determined through, for example, cluster-randomized implementation trials [13] performed after the original clinical trials, as some researchers and public health experts have argued [14, 15, 16, 17]. These have limitations as well, but they could have provided useful additional, real-world data in a relatively controlled setting. However, virtually all countries including The Netherlands chose to roll out the vaccines without such additional trials and assessments. As a consequence we depend almost entirely on retrospective observational studies to draw conclusions about these key endpoints.

An influential retrospective observational study, of a large Dutch cohort based on nation-wide electronic health records including records in the national vaccination database [18], concluded that the vaccines had very high Vaccine Effectiveness (VE) against Covid-19 mortality, of approx. 90% or more, which decreased gradually. These numbers mirror the similarly high VE results in terms of symptomatic infection from the original trials. That study also reported VE numbers with respect to ACM of approx. 50-80%. Similarly high vaccine effectiveness results with respect to ACM have been estimated in most other studies. These include studies based on the same Dutch dataset [19, 20, 21] and studies elsewhere [22, 23]. However, these findings raise questions, as even at its peak Covid-19 deaths never constituted more than 5 to 15% of all deaths [18, 23, 24, 25, 26], even during the main Covid waves, a fact that appears to be incompatible with much larger apparent protection against all types of deaths taken together.

Those same studies [18, 23, 25, 20, 21] acknowledge that the well-known healthy vaccinee effect (HVE), a form of confounding [27], may have affected the results, despite efforts to adjust for it; but it is typically not considered to be a major factor in the results. In addition, other potential systematic biases or confounders, though discussed and acknowledged less often, may affect these types of observational studies. In particular, immortal time bias [28] and vaccination status misclassification bias [29] may play a role. The latter is possible, especially in the Netherlands but also elsewhere, because the Dutch national vaccination database is incomplete due to absence of data sharing consent from some individuals, next to other issues. Importantly, it has recently become clear that the percentage of missing vaccination data is larger than expected, and unevenly distributed among different subgroups [24, 30, 31, 19]. While certainly relevant in the Netherlands, vaccination status misclassification bias may have broader relevance as well, as similar concerns have been raised elsewhere [32, 29, 33].

In this paper we re-analyze the same Dutch national Covid-19 vaccination and health records dataset [18, 34] with a special focus on these bias issues. This dataset has only relatively recently been made available to a wider group of independent researchers. As in [18] and many other studies we use standard survival analysis techniques and stratification into various age and health groups. However, we use matching instead of a posteriori covariate adjustments to attempt to correct as much as possible for HVE-related confounding, in as transparent a way as possible. Also, we employ extra analyses to test, detect, and correct for multiple systematic biases. Due to the presence of and focus on these systematic biases we forgo some of the conventional statistical testing, which in any case

assumes absence of such large systematic biases, in favor of a focus on straightforward presentation, visualization, and interpretation of available and calculated data. Furthermore, for the same reasons we employ multiple additional post-hoc analyses, based on internal and externally sourced additional data and based on quantitative bias analysis (QBA) modeling.

2. Methods

2.1. Study population

Our study population consists of most of the population of the Netherlands, using data available from the Dutch national statistics office CBS. CBS maintains databases of many socio-economic, mortality and health related datasets at the level of the individual citizen, with pseudonymized social security numbers allowing linkage between the datasets. This set of databases is considered to be among the world's best for large-scale real-world observational health research, including Covid-19 related research, due to its nationwide coverage and its broad set of relatively fine-grained health and socio-economic indicators [18, 35, 36]. The data are accessible to researchers upon request and under certain security and privacy conditions, through the "CBS Microdata" remote access research platform [34]. In the Covid pandemic, the datasets were expanded with the CIMS national Covid vaccination database. Since late 2023, CIMS and related datasets have been made available to a wider group researchers than the national government statistics and health institutes themselves, who performed the earlier analyses [18].

Our study cohort includes all registered inhabitants of The Netherlands born between 1930 and 2020. For inclusion in our study cohort, we require that we can reliably track people in the main per-year datasets, from 2018 through 2022, to minimize censoring and missing data problems related to immigration, emigration, homelessness and incomplete health records, etc. (as detailed below and in the Supplementary Information, see Section 5, specifically Appendix A.1). This results in a dataset of 15,915,865 persons.

2.2. Overview of study design

Based on this cohort, we stratify subjects into smaller cohorts for analysis based on age (year of birth). Recipients of long-term care ("WLZ") are analyzed separately, as these constitute particularly vulnerable subgroups. For older groups, we use a 10-year interval for the age cohort (e.g., those born 1930-1940). For the relatively younger groups we use a 20-year interval (e.g., 1980-2000) due to much lower mortality numbers. See the Supplementary Information, Appendix A.1, for precise cohort definitions. These definitions and stratifications are similar to previous studies in the Netherlands using the same dataset [18, 24, 37, 19].

For each cohort, we calculate Covid, non-Covid, and (combined) ACM mortality metrics over time. Based on those we calculate VE metrics using standard methods (as detailed in Appendix A.4). The measurement period encompasses 2021-2023, i.e. the first three years since Covid-19 vaccine roll-out. We primarily use standard survival analysis methods based on Kaplan-Meier (KM) curves, assessing incidence rates of mortality in Covid-19 vaccinated versus Covid-19 unvaccinated persons. We use this method in conjunction with a matching algorithm that aims to correct for HVE-related confounding and immortal time bias.

In addition, we use a secondary approach to visualize and analyze more 'raw' and diverse data based on calendar time. The latter is used to provide complementary data insights into what happens during and after vaccination rounds and to assess seasonality effects. This is useful also because there are strong indications (described below) that standard assumptions of survival analysis such as the proportional hazards assumption and assumptions regarding data integrity and homogeneity of research groups and treatment effects are violated.

In part due to these same violations of the proportional hazards and other assumptions, we do not employ Cox Proportional Hazards regression (most commonly used in survival analysis), but instead calculate and report mortality and VE metrics straightforwardly from the KM data and calendar time-based data. We calculate VE metrics both in terms of relative risk and in terms of absolute risk. For the same reasons of statistical assumptions violation, no standard confidence intervals and statistical significance levels can be calculated, and all calculated and reported VE metrics should be interpreted with great caution, as they do not imply clear causality.

In some analyses, we compare measured mortality rates in the years 2021-2023 to 'normal' (baseline, expected) mortality. Normal mortality for an age cohort is the expected mortality rate for that age cohort based on averaged data from the pre-pandemic years 2016 through 2019 (see Appendix A.1 for the calculation method).

Finally, we perform a number of post-hoc analyses, based on our own and externally sourced additional data and based on quantitative bias analysis (QBA) techniques, in order to better understand, quantify, and (approximately) adjust for the identified biases.

2.3. Vaccination status and follow-up

In order to accurately and completely assess real-world effects of Covid-19 vaccination on short-term and long-term mortality, we start follow-up for vaccinated individuals on the first day of vaccination (dose 1), instead of two weeks after completion of the primary course (typically 2 doses), which has been used in most other studies [2, 18, 24, 38, 20]. This way, we can follow mortality patterns and differences immediately after (first) vaccination, i.e. to also assess very short-term effects. Although we acknowledge that vaccine protection effects such as designed and intended may only emerge two weeks after completion of the full primary Covid vaccination course, this analysis choice renders visible the important data points between dose 1 and two weeks after dose 2; not doing so might introduce or hide biases related to that period [32]. Several of the studies [18, 24, 38, 20] that we compare with in our study do include, typically as secondary analyses, analyses based on similar follow-up immediately after dose 1, making more direct comparison possible.

Follow-up of mortality was done until the end of 2023, allowing some evaluation of long-term effects of vaccination as well. However, with respect to longer-term effects and evaluation, we recognize that earlier Covid-19 waves and variants are very different from later ones and cannot easily be compared in terms of severity and vaccine effectiveness, esp. after the emergence of Omicron, and other effects such as (survival) selection effects play an important role in later mortality. Thus, the primary focus is on shorter-term evaluation, of data in 2021 in particular.

The main analysis focuses on mortality following the primary vaccination round in The Netherlands (which took place in the first seven months of 2021). However, we also include a small extra analysis of data following the first booster round (which took place November 2021 - January 2022; details can be found in Appendix A.11).

As mentioned, vaccination data was obtained from the central Dutch Covid vaccination database (CIMS). This database brings together vaccinations administered by different agencies and organizations. Individuals were asked for consent to be included in this database, which means that a certain percentage of administered vaccinations is missing from the database. National health institutes estimate that data is missing for approx. 7% of vaccinated individuals [18, 19, 37, 21]. However, there are strong indications that missing or incorrect vaccination data is unevenly distributed among subgroups, missing vaccination data is not due to lack of consent alone, and that the percentage of missing data is larger for certain subgroups with higher vulnerability [24, 19]. This could potentially lead to misclassification bias [29]. At the same time, there is no explicit indicator in the CIMS database that someone is unvaccinated. Thus, all people for whom vaccination records are absent in CIMS necessarily have "status unvaccinated"; both in case they are truly unvaccinated and in case they are vaccinated but unregistered. This is an issue affecting also all previous Dutch studies [18, 19, 37, 21, 20]. Thus, the

word "status" in this paper means "classification"; and it can be an inferred classification based on absence of vaccine registration. The opposite situation, people in the Netherlands not being vaccinated but being registered as "vaccinated" does not (or only very rarely) occur, to our knowledge. Therefore, these vaccination status misclassification problems are asymmetrical. We were unable to obtain precise information on the group of people who were vaccinated but whose vaccine registrations are not registered. However, in our analyses below we take into account, and attempt to analyze, this issue explicitly and extensively, in part using secondary data and sources. See Appendix A.2 for more detailed information on vaccination status definitions and the CIMS database, and Appendix A.12 for more detailed and secondary data source information on missing vaccination records and who were affected.

2.4. Causes of death definitions

As described, this study considers both Covid-19 and non-Covid-19 mortality. Covid-19 mortality was defined as Covid-19 having been registered as cause of death on the death certificate by the physician, using standard ICD10 codes (U07.1 and U07.2), as registered in the CBS databases. This is in line with the approach used in earlier analyses, both nationally and internationally [18, 24, 19, 37, 39, 40]. Non-Covid-19 mortality corresponded to any other cause of death. The data available through the CBS research platform do not allow one to accurately account for concomitant diseases or to undertake a more sophisticated analysis regarding primary and secondary causes of death, including potentially post-acute sequelae of SARS-CoV2 infection, a limitation shared with previous studies. Furthermore and partially related, in general there have been concerns about potential overattribution and underattribution to Covid-19 as a cause of death [40], another limitation shared with previous studies. We refer to Appendix A.3 for a more detailed description and discussion of cause of death definitions and the limitations.

2.5. Stratification, matching, and statistical analysis approach

As mentioned, individuals were stratified into cohorts based on age (specifically: using recorded year of birth) and long-term care use (WLZ), similar to previous studies. WLZ indication, including type and level of care, is based on information derived from health insurance data. These individuals constitute mostly institutionalized nursing home senior citizens, but also institutionalized and at home care for individuals with physical or mental disabilities. Conversely, not all nursing home residents, nor many other elderly or vulnerable individuals, have this WLZ indication. Thus, the distinction between WLZ and no WLZ is not clear-cut, but nevertheless it provides an important distinction that can and should be used for stratification. This stratification approach is in line with earlier analyses in the Netherlands [18, 24, 37, 19].

Within each age/WLZ cohort, we use a fixed cohort, "never-treated" (unvaccinated) vs. "ever-treated" (vaccinated) design [41, 42, 43], and we use matching [44] to correct for confounding variables. This is a somewhat different approach than the commonly used approach of Cox Proportional Hazard regression with covariate adjustments, in which treatment typically is a time-varying covariate (used also in [18]). Such an approach based on matching and a never-treated vs. ever-treated design makes sense and has advantages in datasets with large heterogeneity within groups, high likelihood of strongly nonlinear relationships between confounding variables and mortality probability [44], and high levels of cross-over for individuals from status unvaccinated to status vaccinated within a small period of time, all of which we expect here (see Appendix A.5 for additional details). A second reason to use matching is that in some ways it corresponds to a more transparent way to attempt to correct for and investigate confounding than a posteriori covariate adjustment in Cox regression equations [44].

Within an age/WLZ cohort, our matching algorithm matches individuals with status vaccinated and individuals with status unvaccinated based on age, sex, and comorbidity information. We use a 3:1 ratio in matching vaccinated with unvaccinated individuals, because there are typically many

more vaccinated people than unvaccinated people and we wish to minimize data loss for those more rare unvaccinated individuals. As for comorbidity information, we rely primarily on previous-year medication use (i.e. 2020; first dose Covid vaccinations were in 2021). We calculate from that the standard Rx-Risk Comorbidity Index [45], based on medication ATC codes. This a well-known alternative to the Charlson Comorbidity Index (derived from ICD10 diagnosis codes) and others (which we experimented with in preliminary work, see Appendix A.5). For long-term care WLZ recipients, WLZ type and level of care are used for matching, in part because medication use data is more often incomplete for those individuals (who often receive medication through different medical and insurance channels than ordinary citizens). Preliminary analyses showed that on top of these covariates that we used, available socio-economic factors related to country of origin or education, job, and income level (available only at the crude level of the whole household) added relatively little extra predictive information in terms of mortality risk; thus, they were not used.

Furthermore, our matching algorithm requires that an unvaccinated person, who is to be matched with a vaccinated person, is alive on the (first dose) vaccination date of that vaccinated person (who, naturally, is also alive at that point in time). Follow-up for each individual starts on that same date. This realizes the Prescription Time-Distribution Matching (PTDM) requirement [42], designed to prevent immortal time bias [28, 42] and seasonality-related bias [27], which may have affected previous analyses [18]. As matching algorithm, we use the Coarsened Exact Matching (CEM) algorithm [46]. It has been used extensively before and shown to work robustly in many cases, sometimes more so than Propensity Score Matching [47].

Unfortunately, it is impossible to stratify, analyze or match vaccinated and unvaccinated cohorts accurately based on previous SARS-CoV2 infection history, due to the lack of (complete and accurate) testing and infection datasets in the CBS research platform. This is another limitation shared with most previous studies [18, 19] (see also Appendix A.3). We do however perform some analyses based on stratification using comorbidity level (again derived from medication-based Rx-Risk). The Supplementary Information (see 5; specifically Appendix A.5, Appendix A.6, and Appendix A.10) provides additional details on justification, specific implementation, limitations, remaining (selection) biases, as well as intermediate results of the stratification and matching algorithms.

2.6. Calendar time analysis approach

As mentioned above, as a complementary approach next to the KM survival analysis based on matched cohorts, we also perform additional analyses and visualizations using calendar time-based data, utilizing all available and diverse population data within an age cohort. This uses unmatched and unadjusted data, but similarly stratified by age and long-term care use (WLZ). This approach allows us to better assess the effects, over time, of different subgroups being vaccinated in different months, seasonal effects, et cetera. Furthermore, it allows us to assess time-varying mortality patterns alongside various other data to heuristically inspect likely relationships. These other data include vaccination dose and vaccination rate data, baseline mortality data, and detailed causes of death data. And this allows us to study, in a complementary way, the observed and potential residual confounders and other issues making standard KM survival analysis challenging. It allows us to do this without the specific data transformations and potential biases that might have been introduced by the KM analysis framework and our specific matching algorithm.

Finally, because these calendar time analyses are based on all available population data, these complementary analyses also allow analysis of overall, total population level mortality effects, i.e. to observe increases and decreases in total population mortality (excess mortality) compared to normal baseline levels. Similar calendar time analyses were done using our smaller matched cohorts as well, showing qualitatively similar results (see Appendix A.9); but this loses the perspective on the overall population level data patterns.

2.7. Additional statistics related to the identified biases

As an additional, post-hoc analysis following the main analyses, we calculate, collect, and report additional descriptive statistics, based both on our own data and on external data. These data and statistics are related to both the integrity of the vaccination dataset and to secondary data on individuals' high vulnerability (in terms of health, extreme frailty, and short-term mortality risk). They are intended to provide extra, secondary information on the suspected biases/artifacts suspected and discovered in the main analyses, i.e. the issues of residual HVE-related confounding and vaccination status misclassification bias. These analysis methods and data are described in more detail below and in the Supplementary Information, Appendix A.12.

2.8. Quantitative bias analysis

Finally, we use post-hoc quantitative bias analysis (QBA) in an effort to better understand, quantify, and (approximately) adjust for the suspected/discovered biases. QBA can be understood as a form of sensitivity analysis particularly called for and well-suited in cases where the evidence suggests large residual systematic (non-random) errors [48, 49], as is the case in this study. This includes less "traditional" and less common issues (relevant here) that are related to handling of missing data and exposure/treatment misclassification, which cannot easily be analyzed in other ways. We perform QBA in two different but complementary ways (methods are explained in more detail in the Supplementary Material, Appendix A.13 and Appendix A.14). Both of them are imperfect and mostly exploratory, intended to complement the primary analyses and each other. Although grounded in empirical data, as always with QBA they are dependent on and sensitive to model choices and assumptions.

The first approach focuses on and models the potential underlying mechanisms driving the biases, similar to many other QBA efforts [48, 49]. It uses plausible, mathematically modeled mechanisms and plausible parameters (derived from our study's data) to simulate the biases probabilistically, in an effort to simulate and thereby understand the combined processes that could lead to the observed Kaplan-Meier survival data. The process starts with a simulated population of vaccinated and unvaccinated individuals at start of follow-up. It then simulates what would be expected to happen over time from start of follow-up onwards, in terms of expected mortality for groups with real and apparent vaccination status "vaccinated" and "unvaccinated" (which includes vaccinated but unregistered individuals). It calculates these simulated data given modeled assumptions and parameters representing expected underlying and varying vulnerability and mortality patterns, together with assumed biases. In essence, this approach models the bias factors that, based on the data and analyses above, we expect to play a role; and attempts to assess how large each factor needs to be to somewhat realistically produce (mimic) the measured KM survival data. See Appendix A.13 for more details.

The second QBA approach, instead, largely ignores the potential underlying factors or mechanisms driving the biases, modeled in the first QBA approach. In contrast, it is numerical and purely data-driven. It corresponds to a heuristic *negative control data*-based systematic bias correction. It has strong similarities to several other QBA efforts that use negative control data in a similar way, both in the Covid domain and outside it [33, 50, 51, 52, 53, 49]. It uses the assumption that the Covid vaccines should not meaningfully reduce risk for causes of death other than Covid (the negative control). This assumption (or something very similar) is used also elsewhere in this paper, in other Covid studies, and in studies of other vaccines and viruses like influenza [25, 26, 54, 27]. In other words, non-Covid mortality data, and observed differences therein between vaccinated and unvaccinated groups, constitute reasonable negative control data, which may be exploited quantitatively in QBA. Based on this starting point, the method estimates a time-varying multiplicative correction factor that effectively balances non-Covid mortality between the two groups. This corresponds to a simple normalization or scaling approach. It then applies this correction factor to the measured all-cause mortality rates per day, which are accordingly normalized (and also Covid mortality rates, as a secondary target). Based on this, finally, bias-adjusted KM survival curves can be calculated. This approach has the attractive

property that it corresponds to a pragmatic, heuristic, data-driven, net correction on the all-cause (or Covid) KM data, without needing to disentangle the composite sources of bias or explicitly model the causal mechanisms. It only assumes that the net residual biases (HVE and vaccination status misclassification together) affect all-cause and Covid mortality proportionally to their observed effect on non-Covid mortality, the observed "pseudo-effect" on the negative control data, as it sometimes called [33]. See Appendix A.14 for more details.

3. Results and discussion of results

3.1. Baseline characteristics of the study population, before and after matching

Results. Figure 1 depicts a consort flow diagram showing the overall study population and the exclusions. Appendix A.1 in the Supplementary Information (see 5) describes these inclusion and exclusion criteria in detail. Demographic and other baseline characteristics of the study population are shown in Table 1. (This table includes also the small 2000-2020 group with WLZ long-term care, that is excluded in other analyses due to very low number of deaths in that group.) Standardized Mean Difference (SMD) calculations [55] are also included where appropriate, to assess class imbalances. SMD values, which are similar to and can be interpreted as a type of "effect size", larger than roughly 0.2 (in absolute terms) indicate a relatively large class imbalance.

In Table 1, three "ethnicity classes" are distinguished based on place of birth of the individuals and their parents. Ethnicity class A corresponds to individuals who were born in the Netherlands and their parents as well. Class B means the individual was born in Netherlands, but at least one parent was born abroad. Class C means individuals were themselves born abroad. Income level, education level, and employment level are directly based on corresponding metrics available through the CBS microdata environment, ranging between -1 and +1, with 0 representing a median value. These are only available at the household level, i.e. not at the individual level, meaning that they are imprecise.

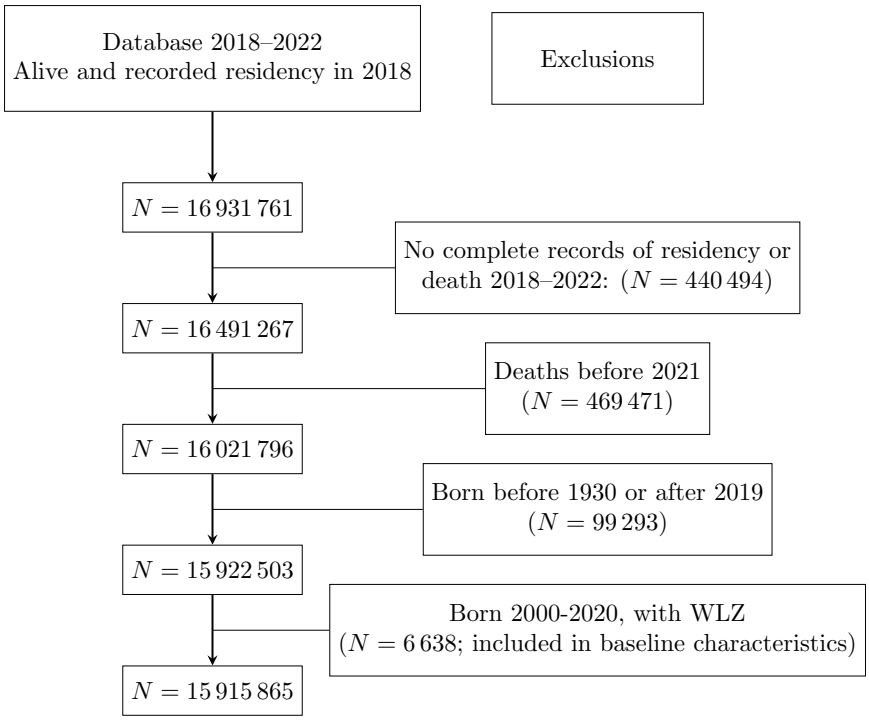


Figure 1: Consort Flow Diagram showing the overall study population and the exclusions.

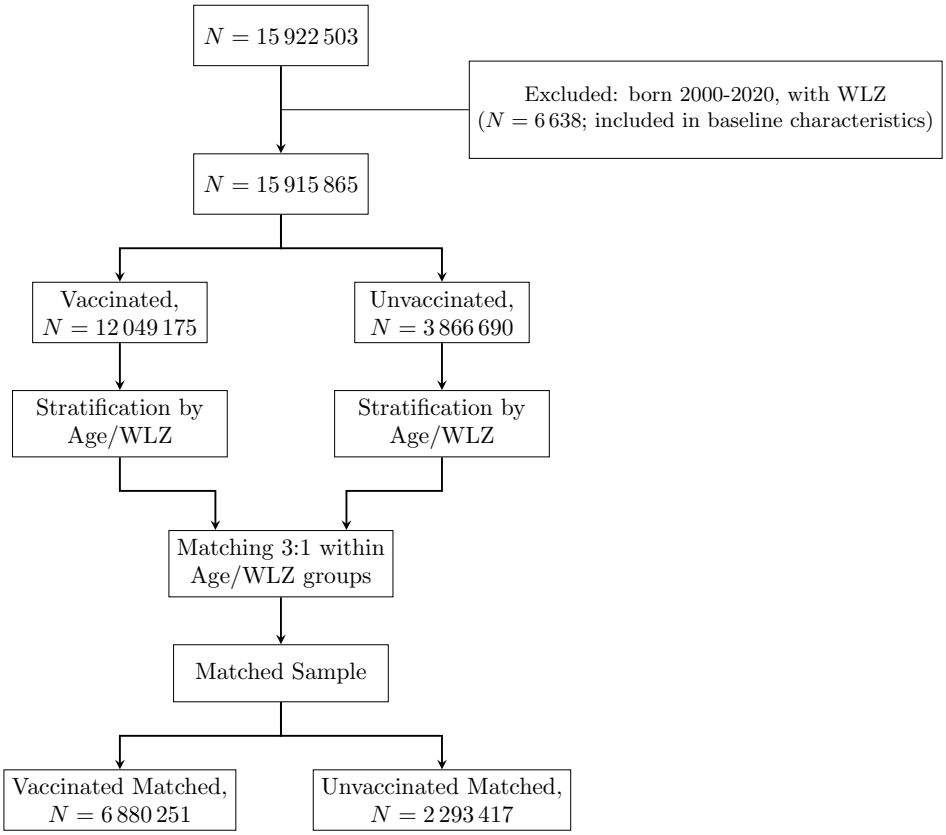


Figure 2: Consort Flow Diagram showing the stratification and matching process.

Figure 2 depicts a consort flow diagram showing the stratification and matching process based on the overall study population (described in Section 2.5), resulting in the cohorts used in most analyses. As described in Section 2.5, the main factors affecting mortality risk, namely age, WLZ indication (and if so, type and level), sex, and the Rx-Risk comorbidity metric were used in stratification and matching. Appendix A.6 in the Supplementary Information (5) shows characteristics of the age/WLZ cohorts after stratification and matching. These also show the resolved and limited remaining imbalances of the main baseline variables.

Discussion. As shown in Table 1 (age group, % in age group), vaccination rates (defined here as: at least one dose, see Section 2.3) are high, around 90%, in all age groups except the youngest groups. Note that this vaccination rate is the percentage as calculated directly by linking the pseudonymized personal IDs in the CBS microdata population and health datasets to those in the national CIMS Covid-19 vaccination database. As described in Section 2.3, not all actual vaccinations were registered. This means that the percentages vaccinated as provided in the table are an underestimation. In the youngest group, born 2000-2020, only the relatively older ones (teenagers) were actively invited for Covid vaccination, and relatively many of those invited declined, making vaccination rate relatively low there. Overall, in the whole study population, the ratio of individuals with status "vaccinated" vs. those with status "unvaccinated" is roughly 3:1. Most individuals who received dose 1 of a two-dose primary series vaccination also received dose 2 (results shown later).

There is little to no imbalance in terms of male/female ratio. Naturally, for reasons described above, the vaccinated group is, on average, older. Related to this substantial age imbalance (i.e. before age stratification), on average the vaccinated group has higher comorbidity as measured by previous year medication use-based Rx-Risk. Likewise, WLZ (long-term care) use is higher in the vaccinated group.

Table 1: Demographic and baseline characteristics of the study population before matching, by vaccination status, with Standardized Mean Difference (SMD) where appropriate.

		Status Vaccinated	Status Unvaccinated	SMD
Number of people	N (%)	12054368 (75.71)	3868135 (24.29)	
Age	mean (std.dev.)	49.7 (20.4)	29.4 (20.8)	0.984
Age group	N (% in status group)			
1930-1940		572270 (4.75)	57642 (1.49)	0.188
1940-1950		1417247 (11.76)	107375 (2.78)	0.351
1950-1960		1896581 (15.73)	176121 (4.55)	0.377
1960-1980		3889529 (32.27)	704352 (18.21)	0.328
1980-2000		2847376 (23.62)	1115454 (28.84)	-0.119
2000-2020		1431365 (11.87)	1707191 (44.13)	-0.770
Age group	N (% in age group)			
1930-1940		572270 (90.85)	57642 (9.15)	
1940-1950		1417247 (92.96)	107375 (7.04)	
1950-1960		1896581 (91.50)	176121 (8.50)	
1960-1980		3889529 (84.67)	704352 (15.33)	
1980-2000		2847376 (71.85)	1115454 (28.15)	
2000-2020		1431365 (45.61)	1707191 (54.39)	
Male	N (%)	5929515 (49.2)	1968794 (50.9)	-0.034
Ethnicity class A	N (%)	9831788 (81.56)	2509387 (64.87)	0.384
Ethnicity class B	N (%)	955892 (7.93)	808339 (20.90)	-0.376
Ethnicity class C	N (%)	1266688 (10.51)	550409 (14.23)	-0.113
Income level	mean (std.dev.)	0.100 (0.31)	-0.036 (0.36)	0.402
Education level	mean (std.dev.)	0.058 (0.27)	0.017 (0.28)	0.146
Employment level	mean (std.dev.)	0.080 (0.28)	0.040 (0.37)	0.120
WLZ (long-term care)	N (%)	174309 (1.45)	28325 (0.73)	0.069
Rx-Risk Comorbidity	mean (std.dev.)	1.641 (3.44)	0.880 (2.68)	0.247

The unvaccinated group has a higher proportion of individuals of non-native Dutch origin. This is in part related to the age imbalance: migration has been higher in recent decades compared to earlier. However, this imbalance is likely also due to other factors affecting differential vaccine uptake in different socio-economic and ethnic groups [5, 27, 37]. In line with these insights on socio-economic factors, in our study population the vaccinated group on average had higher education level, income level, and job participation rate.

As described, only the main factors affecting mortality risk, namely age, WLZ indication (and type and level), sex, and the Rx-Risk comorbidity metric were used in stratification and matching. Appendix A.6 (in the Supplementary Information, 5) shows that, after matching, in the stratified and matched cohorts there are no major imbalances in these main baseline characteristics (Tables A.1 and A.2). A minor exception is a small imbalance, among WLZ recipients, in terms of Rx-Risk comorbidity based on previous year medication use data; as described, medication use data is incomplete for that group and is not (could not reliably be) used for matching there. As can also be seen in Appendix A.6, among WLZ long-term care recipients, in the older groups the most common WLZ type is 1, corresponding to nursing homes and assisted living facilities. In the younger groups, the most common WLZ type is 4, corresponding to care for individuals with congenital intellectual disabilities. See Appendix A.6 and Appendix A.5 in the Supplementary Information for these and extra data and details on this and other results of the matching process.

3.2. Kaplan-Meier survival analysis for matched cohorts

Results. Utilizing the methods described in Section 2 and the resulting stratified and matched cohorts described in Section 3.1, Kaplan-Meier survival data for matched vaccinated and unvaccinated cohorts were calculated for all age/WLZ cohorts. Figure 3 shows results for age cohort 1940-1950, no WLZ (no institutionalized long-term care). We focus on this group for presentation purposes, and for the following reasons: it is a relatively old group for which Covid risks are relatively high and vaccination was expected to be a substantial factor in helping to survive Covid; but not the very oldest group in our study, which is a lot smaller and which has a lot more natural depletion due to very old age. Furthermore, the main data patterns are similar to what we see in other age/WLZ groups and therefore typical. Results for other age/WLZ groups are shown below and in the Supplementary Information.

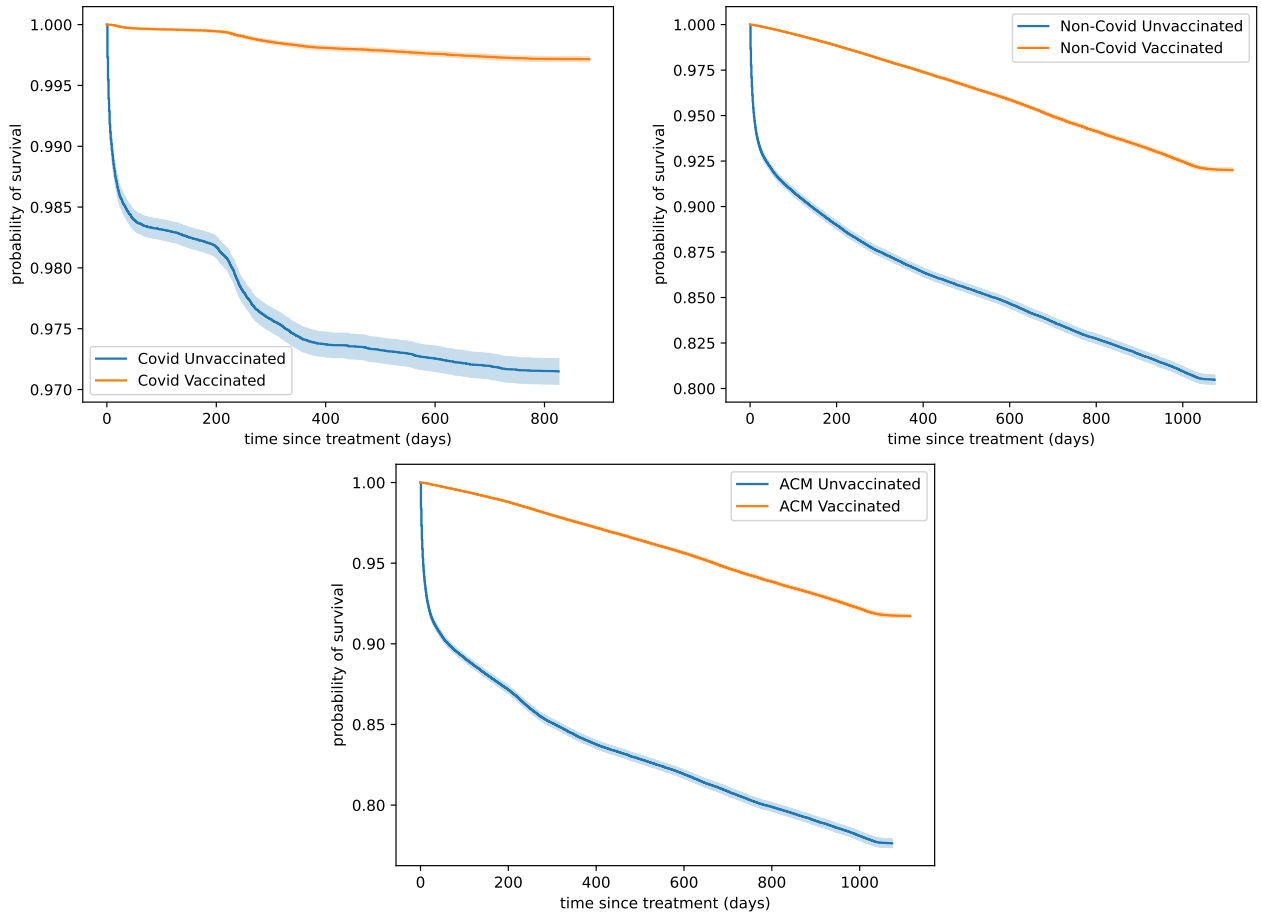


Figure 3: Survival curves of vaccinated and unvaccinated individuals of age cohort 1940-1950, no WLZ use, for mortality from Covid (top left); cause of death other than Covid-19 (top right); and All-Cause Mortality (bottom), respectively. Note the different scales on the y-axis.

Results for this age group, in Figure 3, are shown for mortality from Covid (top left); cause of death other than Covid (top right); and ACM (bottom), respectively. The curves run from various starting dates in vaccination round 1 (first six months of 2021, for almost everyone) to the end of 2023, corresponding to over 1000 days. Note the different scales on the y-axis: Covid mortality represents only a small proportion of total mortality, even for this relatively old group.

Summary results for other age/WLZ groups are shown in Tables 2 and 3, and in the Supplementary Information, Section Appendix A.7. Table 2 shows survival percentages (in terms of all-cause mortality,

Table 2: Survival percentages (ACM) of the unvaccinated (S_u) and vaccinated (S_v) cohorts, as derived directly from the KM survival data determined for all stratified and matched age/WLZ cohorts. Data are shown for 28 days (4 weeks) since start of follow-up; 365 days (1 year); and 1000 days (almost the end of the entire measurement period). These values should interpreted with great caution, as they are subject to large systematic biases or artifacts; see text for explanation.

age cohort	WLZ	S_u^{28}	S_u^{365}	S_u^{1000}	S_v^{28}	S_v^{365}	S_v^{1000}
1930-1940	0	88.674	68.604	54.280	99.644	93.000	78.248
1940-1950	0	91.412	84.170	78.089	99.862	97.469	92.186
1950-1960	0	96.578	93.854	91.272	99.956	99.111	97.220
1960-1980	0	99.512	98.848	98.231	99.991	99.785	99.335
1980-2000	0	99.962	99.875	99.781	99.998	99.961	99.890
2000-2020	0	99.985	99.950	99.913	99.999	99.982	99.951
1930-1940	1	71.510	34.040	17.160	97.970	72.105	37.098
1940-1950	1	78.348	44.747	29.190	98.672	80.380	52.419
1950-1960	1	86.982	64.891	51.800	99.360	90.171	72.781
1960-1980	1	96.300	86.820	81.451	99.938	97.039	90.960
1980-2000	1	99.700	98.655	97.970	99.984	99.640	98.983

Table 3: *Apparent* Vaccine Effectiveness (VE) values (ACM) as calculated directly from the KM survival data for all age/WLZ cohorts. These values should interpreted with great caution, as they are subject to large systematic biases or artifacts; see text for explanation.

age cohort	WLZ	VE_{RRR}^{0-28}	VE_{RRR}^{28-365}	$VE_{RRR}^{365-1000}$	VE_{ARR}^{0-28}	VE_{ARR}^{28-365}	$VE_{ARR}^{365-1000}$
1930-1940	0	96.9	66.9	−3.0	11.0	13.4	−0.4
1940-1950	0	98.4	67.0	13.1	8.4	4.8	0.8
1950-1960	0	98.7	69.0	26.8	3.4	1.9	0.7
1960-1980	0	98.1	69.0	27.0	0.5	0.5	0.2
1980-2000	0	93.9	57.3	24.9	0.0	0.0	0.0
2000-2020	0	95.1	49.2	16.6	0.0	0.0	0.0
1930-1940	1	92.9	31.0	−107.4	26.5	11.6	−18.1
1940-1950	1	93.9	45.6	−79.7	20.3	15.3	−12.4
1950-1960	1	95.1	58.4	−32.8	12.4	12.9	−4.3
1960-1980	1	98.3	69.4	−13.2	3.6	6.6	−0.7
1980-2000	1	94.7	80.1	4.1	0.3	1.4	0.0

ACM) of the unvaccinated (S_u) and vaccinated (S_v) cohorts at different moments (numbers of days) since start of follow-up. Data are shown for 28 days (4 weeks) since start of follow-up; 365 days (1 year); and 1000 days (almost the end of the entire measurement period). These are somewhat unusual, "raw" data to report directly, reported here to allow assessments of plausibility and their role in calculation of vaccine effectiveness (or equivalently, hazard ratios or risk ratios).

Table 3, in turn, shows *apparent* VE values (w.r.t. ACM) as calculated directly from these KM survival percentages, for the same periods defined by these follow-up moments, both in terms of relative terms and in absolute terms. E.g. VE_{RRR}^{28-365} for age cohort 1930-1940, WLZ 0, shows the calculated Vaccine Effectiveness in terms of Relative Risk Reduction (see Appendix A.4), for this age group,

no WLZ (no long-term care), based on the relative mortality rates of the matched unvaccinated and vaccinated subgroups, between day 28 and day 365; as derived directed from the survival percentages shown in Table 2. All these calculated values should be interpreted with great caution, as they are likely subject to, and in fact suggest and highlight, large systematic biases or data artifacts (as described throughout this paper). Thus, (again) they are included both for reasons of completeness and transparency and to allow the reader to assess (im)plausibility of the results.

Discussion. What stands out in Figure 3 is the very high mortality rate of the unvaccinated with respect to both Covid and (especially) non-Covid and ACM in the first period – especially the first 4 weeks – after start of follow-up (i.e. during and right after the first Covid vaccination round), compared to the vaccinated. This higher than normal mortality rate for the unvaccinated is remarkable especially because for the unvaccinated no intervention takes place in that period. This is, therefore, an anomalous finding which suggests a methodological or data artifact. In contrast, the vaccinated show a much more gradual mortality pattern, more in line with what one would normally expect. This graph also shows that the proportional hazards assumption of standard survival analysis (and specifically methods like Cox Proportional Hazards regression) is violated: in the first weeks after start of follow-up, the differences between vaccinated and unvaccinated people, in terms of mortality probability per day, are much larger than later.

When taken at face value, these survival data say that the unvaccinated in this group have a risk of dying of any cause of death (ACM), in the first 4 weeks after start of follow-up, of as much as 8.59% (see also Table 2). This is extremely and unrealistically high, even for this old cohort. At the same time, for vaccinated individuals, the risk of dying is only 0.14%. Thus, in terms of apparent vaccine protection, in that short 4-week period, the Covid vaccine would provide an absolute risk reduction (VE_{ARR}) for ACM of 8.45%, and a relative risk reduction (VE_{RRR}) of 98.4%, for all causes of death combined. Such a high protective effect immediately after the first dose, especially for all causes of death other than Covid-19 itself, is biologically implausible and instead suggests substantial remaining bias or another type of artifact.

We furthermore observe, in the top two plots in Figure 3 (note the different scales), that in terms of *absolute* risks and risk reduction (VE_{ARR}), these data would suggest that Covid-19 vaccine protection against non-Covid death is much higher than against Covid death itself. This is a somewhat different indicator of systematic bias. This sanity check is a form of negative controls analysis, which has been advised as a standard check in observational epidemiological studies [54].

Moreover, we observe that as time progresses from start of follow-up the survival curves' slopes, i.e. the death rate per day (which should be related to vaccine protection), quickly become increasingly similar for vaccinated and unvaccinated individuals. When computing time-varying VE from such increasingly similar relative mortality rates, calculated VE against ACM rapidly decreases. For example, when calculating vaccine effectiveness against ACM directly from the KM data (see Tables 2 and 3) for the follow-up period of 28 days until 365 days, apparent VE_{RRR} is 67.0%; and for the period of 365 days until 1000 days (almost the end of the measurement period), apparent VE_{RRR} is 13.1%.

For the same age group but for individuals with WLZ use (long-term care recipients), we see essentially the same patterns, but even more extreme, with even higher and unrealistic overall death rates for the people with status "unvaccinated". See Tables 2 and 3 and the extra data and figures in Section Appendix A.7 of the Supplementary Information (5). When calculating VE against ACM in this way for this group, for the period of 0 to 28 days we calculate apparent VE_{RRR} as 93.9%; for the period of 28 days until 365 days, VE_{RRR} is 45.6%; and for the period of 365 days until 1000 days, VE_{RRR} even becomes strongly negative, namely -79.7%. Analogous to the conclusions regarding the other calculated VE metrics determined directly from the data, this does not imply real or clear evidence for actual negative vaccine effectiveness; again due to likely artifacts. In this case, given the

low numbers of unvaccinated people involved, and high death and attrition rates overall, over time possibly a strong selection effect occurs, with a small subgroup of elderly, unvaccinated and possibly unusually healthy individuals surviving others.

For younger age groups, we see similar patterns, but naturally with lower overall death rates. See Tables 2 and 3 and the Supplementary Data (Appendix A.7) for additional data and example figures. Thus, we always see a similar pattern of unrealistic, initially very high but rapidly decreasing and sometimes negative, VE against ACM, when we take the KM data at face value and calculate VE from that. All this suggests that similar, and similarly large, artifacts/biases play a role in the data and analysis for all age cohorts.

A potential, straightforward explanation for these findings is the existence of substantial HVE-related residual confounding; despite stratification and matching on age, comorbidity and other information. Besides impacting the KM data and derived VE for Non-Covid and All Cause Mortality, the likely biases probably affect Covid mortality rates and VE calculations as well. After all, the Covid and non-Covid KM curves show very similar patterns and appear highly correlated.

On a methodological level, the results strongly suggest that standard KM survival analysis-derived VE metrics (or equivalently Hazard Ratios, HR), and standard statistical significance tests and confidence intervals, cannot be reliably computed due to clear violations of several underlying assumptions for survival analysis. These assumptions require absence of such large systematic biases and assume proportional hazards and sufficient homogeneity of subgroups [56]; all of which are not the case here. These issues likely affect also previous studies based on the same dataset [18, 19, 20]).

3.3. Differentiation by comorbidity score

Results. The KM data presented above could be interpreted as an indication that the matching algorithm is ineffective, and in particular that the comorbidity data we use for matching contains only poor predictive information regarding a priori health and mortality risk, such that matching on that data does not correct well for health differences (leading to large HVE bias). To investigate this, we do a subgroup analysis utilizing stratification based on comorbidity data.

Figure 4 again shows survival curves for the same 1940-1950 birth year cohort (no WLZ, ACM), but now for subgroups differentiated by comorbidity score Rx-Risk, the main metric used for matching. Note that as described in section 2.5, our Rx-Risk metric is derived from previous year medication use. We differentiate a low Rx-Risk scores subgroup ("R0": Rx-Risk < 1) and high Rx-Risk scores subgroup ("R1": Rx-Risk ≥ 1), respectively.

Discussion. We conclude from Figure 4 that the Rx-Risk comorbidty metric does contain significant information that predicts mortality well. People with high Rx-Risk scores are much more likely to die (both per day and cumulatively) than those with low Rx-Risk scores; and this is true for both unvaccinated and vaccinated people. We also observe that, apart from the first weeks after start of follow-up, the instantaneous hazard per day (slopes of the KM curves) for both vaccinated and unvaccinated individuals of the same Rx-Risk subgroup (low or high) becomes very similar.

Thus, Rx-Risk appears to be a reasonable measure for representing comorbidity and a priori health and risk of dying, and therefore for matching. However, there are again, or still, strong indications of substantial residual bias, pertaining mostly to the first few weeks after start of follow-up. Unique to the group with status unvaccinated, and for both the low comorbidity subgroup and the high comorbidity subgroup, there is a sharp dip in survival indicating unusually high death rates, in those first weeks of follow-up. Besides residual HVE, the extreme size and time-varying nature of this effect, despite careful matching and even when stratifying based on comorbidity level, suggests an additional artifact, including potentially vaccination status misclassification bias, concentrated on the first period since start of follow-up (the start and follow-up of the vaccination round). The next section and section 3.5 discuss these residual HVE and misclassification biases in more detail.

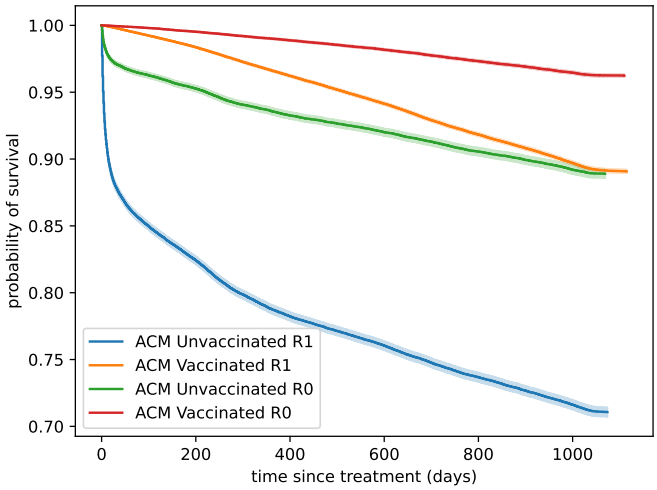


Figure 4: Survival curves for the 1940-1950 age cohort (no WLZ, ACM); with separate subgroups based on low Rx-Risk comorbidity score (R0) versus high Rx-Risk comorbidity score (R1).

3.4. Calendar time analysis

Results. As described in section 2.6, calendar time-based analyses allow us to look at the data in a complementary and more comprehensive way, when compared to the KM survival analyses of the previous sections. And they allow us to investigate and visualize relationships, over time, with multiple, various data sources, as well as seasonality effects.

Figure 5 shows measured and calculated data for the same age cohort as before, 1940-1950, no WLZ use. Each data point corresponds to data aggregated within a 4-week calendar time period. The horizontal axis of each subplot corresponds to week number, increasing from 1 (1st week of 2021) to the end of the entire analysis period, the end of 2023 (i.e., week number 52 is the end of 2021, and 156 is the end of 2023). The subplots contain various and diverse data. This includes, in row A, number of (registered) vaccinations (doses 1, 2, 3), total vaccination rate of the cohort, and numbers of vaccinated and unvaccinated individuals over time. Vaccination rate is, as before, based on having received dose 1. Dose 3 is the last dose included in all graphs and analyses; for most people this corresponds to the first booster vaccination. In row B, overall mortality as well as mortality broken down by vaccination status are shown. In the third and fourth rows (C and D) VE, calculated both in terms of Relative Risk Reduction (VE_{RRR}) and in terms of Absolute Risk Reduction (VE_{ARR}), for Covid-19, non-Covid-19, and ACM mortality are shown. These are time-varying and period-specific VE estimates, to assess changes in these estimates over time. Row D also shows registered causes of death. The mortality subplots (row B) also report 'normal' mortality; as detailed in Section 2.2 and Appendix A.1 this is the baseline mortality for that age cohort as calculated based on averaged data from pre-pandemic years. In those baseline calculations, all individuals regardless of WLZ use (which typically denotes high vulnerability) were included. This is why realized mortality for the no-WLZ cohort shown in Figure 5 is slighter lower than that aggregated baseline (for the WLZ cohort it is higher). Realized mortality over the three years naturally has a slightly upward trend: this relatively old 1940-1950 cohort gets even older in the years 2021 through 2023, and mortality rate for remaining, still-alive individuals per 4 weeks increases correspondingly.

Figure 6 shows the same data, but for WLZ (long-term care) recipients. As expected, mortality rates are much higher in this extremely vulnerable group, compared to the no WLZ group and to the 'normal' rate for the entire age group. Similar calendar time analyses were done using the matched data as well (as in the Kaplan-Meier analyses), showing qualitatively similar results (see Appendix A.9).

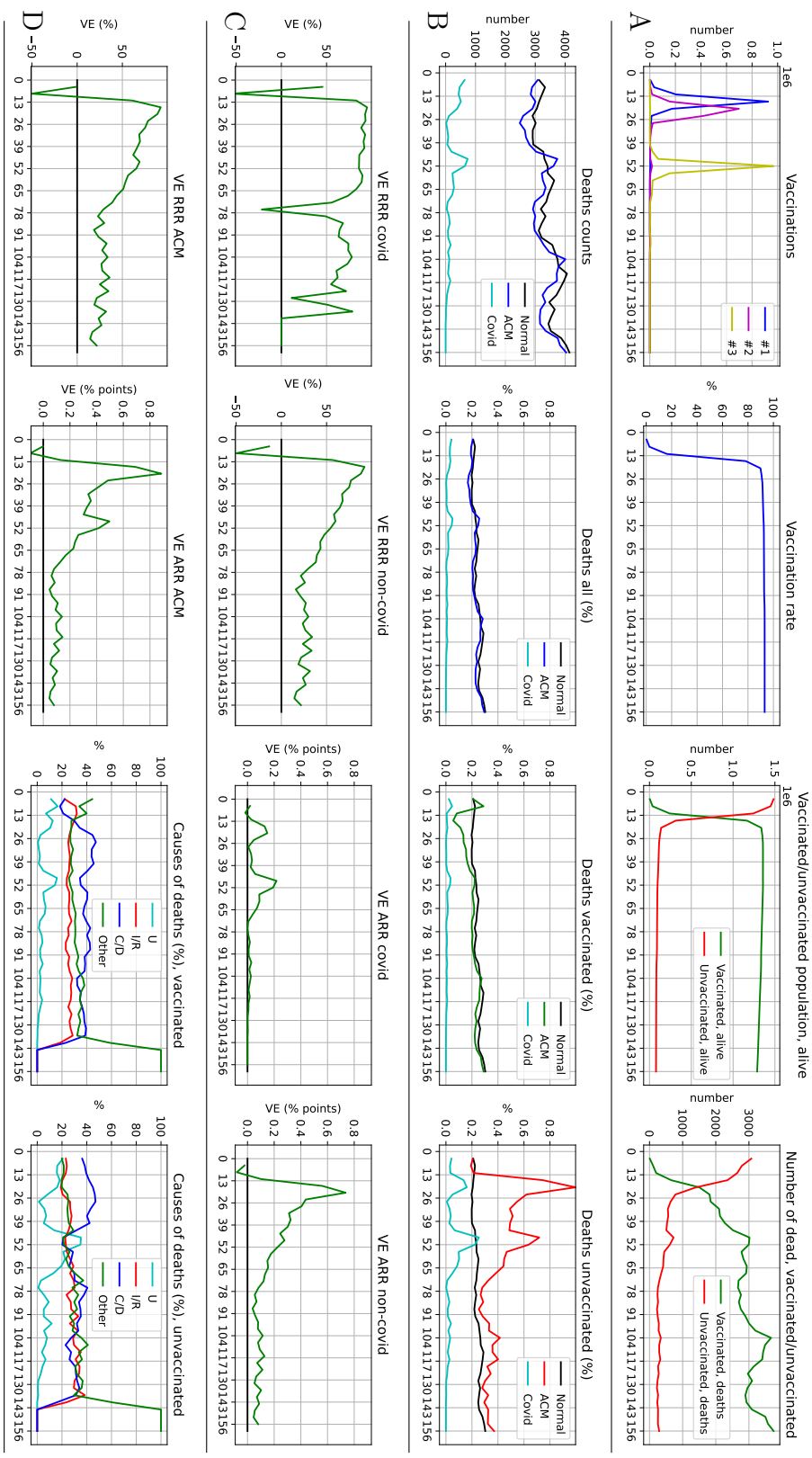


Figure 5: Per-4-weeks calendar time data of age cohort 1940-1950 (unmatched), no WLZ, 2021-2023. Horizontal axes always denote week number since 1 Jan 2021. See text for detailed explanation.

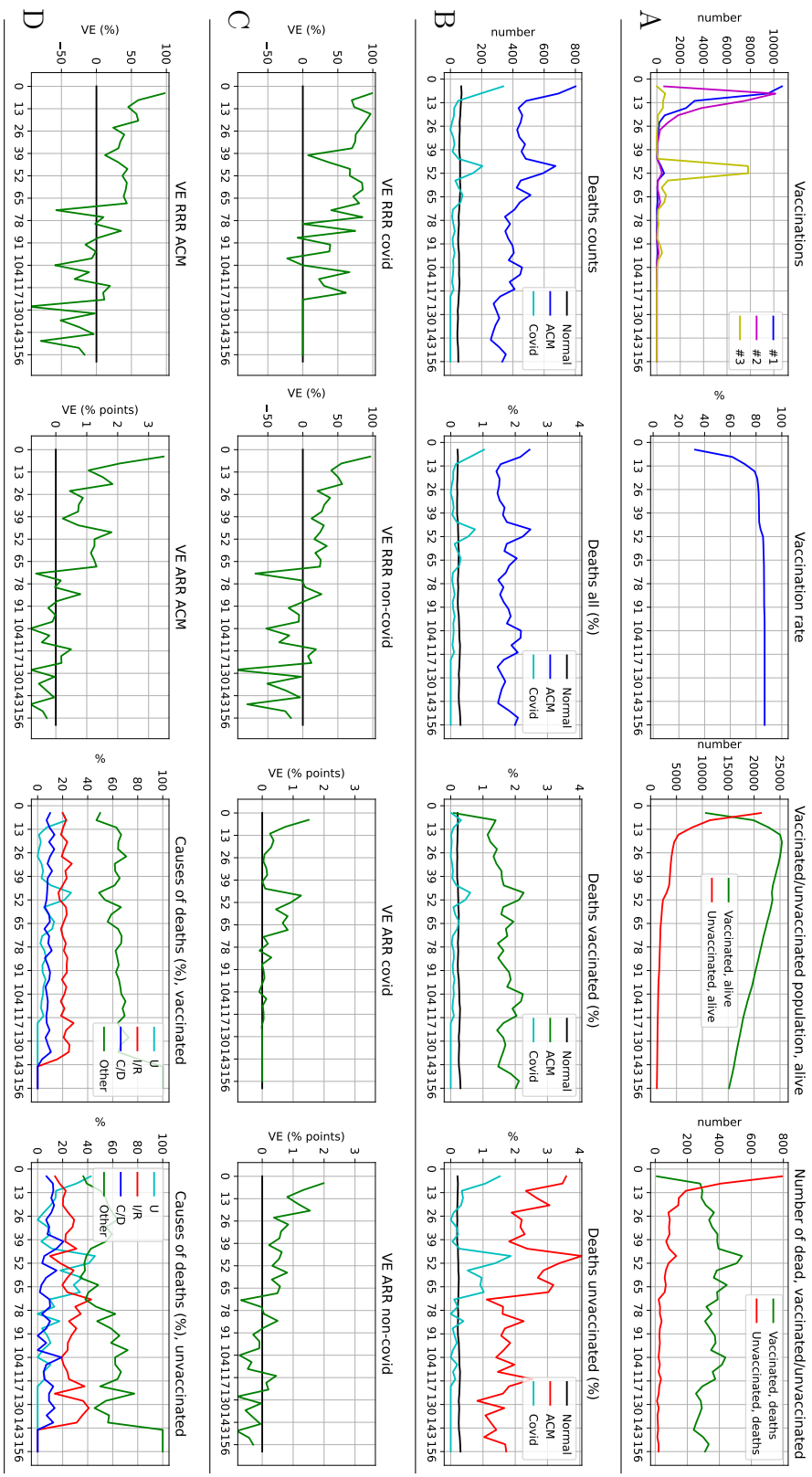


Figure 6: Per-4-weeks calendar time data of age cohort 1940-1950 (unmatched), with WLZ (long-term care), 2021-2023. Horizontal axes always denote week number since 1 Jan 2021. See text for detailed explanation.

Discussion. Similar to before, a striking observation in Figure 5 is the greatly increased mortality among unvaccinated individuals during the first vaccination round, of spring 2021. This is consistent with the mortality pattern observed in the KM survival curves in Figure 3, based on the same dataset (apart from matching, etc.). Only a small part of this mortality can be attributed to Covid-19 (light blue in the subplots of row B, code U in the bottom-right subplots of row D): Covid mortality is somewhat elevated, but constitutes only a small fraction of total mortality. The two subplots in the lower right show the percentages of reported causes of death among the vaccinated and unvaccinated, respectively. See Appendix A.3 for the causes of death codes. (In the causes of death dataset we used, causes of death were not complete by the end of 2023 at the time of analysis, hence in that (last) period the cause of death is always other/unknown.)

By far the most frequent cause of death in the period(s) in the first half of 2021 with greatly increased mortality among unvaccinated people is *cancer* (code C/D). Its relative rate increased in that period, when compared to the more 'normal' looking pattern of causes of death seen in 2022 and 2023, and when compared to vaccinated people. This is yet another indication that there we are dealing with a special situation or subgroup; namely, an unvaccinated subgroup that was already very vulnerable, with many already at death's door, in large part due to cancer.

If we look at the mortality rates of that particular unvaccinated subgroup, we see that at its peak (coinciding with the peak of vaccination round 1), mortality per 28 days of calendar time is up to 500% higher than normal, baseline mortality for this age group. This increase in unvaccinated mortality coincides with a corresponding, temporary *decrease* in mortality among the vaccinated: lower than normal mortality. These findings again reinforce the hypothesis that significant confounders remain. All this is in line with the hypothesized short-term HVE bias and also potentially vaccine status misclassification bias.

Interestingly, at the start of 2021, when in The Netherlands the most vulnerable elderly people were vaccinated for the first time, we observe a small peak in vaccinated mortality – perhaps due to very transient *negative* HVE, or due to other factors. For a short time, this leads to negative apparent VE, as shown in the bottom two rows (C and D). As described above, due to the suspected biases and artifacts, in any case calculated VE values, positive or negative, should be regarded with suspicion, and clear causal conclusions are impossible to make.

Unvaccinated mortality remains very high compared to normal baseline levels during the summer, when respiratory viruses typically play only a small role in mortality. Such a seasonality-related analysis is another form of negative controls analysis, advised in e.g. [54]. In the influenza literature such analyses have cast doubt on calculated effectiveness results for flu vaccines [27]. Mortality rate peaks again in the winter of 2021/2022, together with elevated Covid-19 mortality and the first Covid-19 booster vaccination round.

If we ignore the high likelihood of the hypothesized artifacts/biases underlying these mortality data and take these data at face value, we obtain very high but gradually decreasing time-varying VE values over time; see the bottom two rows C and D. The corresponding VE values and time-dependent patterns are quite similar to those of previous studies using the same data [18, 19] or data from other countries [22, 23]. Again, our analyses taken together strongly suggest that much if not all of the apparent vaccine protection against ACM in these data is the result of the described artifacts.

For the long-term care WLZ group (Figure 6), the patterns are very similar. This vulnerable and relatively small group was mostly vaccinated earlier in the year, in the first few months of 2021, and the patterns shift accordingly. In this dataset the registered vaccination rate for this vulnerable group appears to be lower compared to the overall rates for the age group. This is in contrast with what has previously been reported [37] for vulnerable groups, and what we would expect to see in a long-term care population. Furthermore, first-dose vaccination rate notably increases much later in the year during the booster round in the winter of 2021/2022: relatively many people appear to receive their first dose only then. For this elderly, long-term care group, which we have tracked from the beginning

of 2021, we consider this unlikely. We consider this supporting evidence for the hypothesis that during the first vaccination round, relatively many vaccinations were not properly registered, especially among elderly and vulnerable subgroups in the population. Importantly, and as hypothesized, this could lead to non-negligible misclassification bias.

For both the no-WLZ group and the WLZ group, *overall* mortality (row B, deaths counts and deaths all percentages) of vaccinated and unvaccinated individuals combined is elevated during vaccination rounds (doses 1, 2, 3), concurrent with simultaneous elevated Covid-19 mortality. This is the case despite the fact that for these groups vaccination occurred at different periods in the year. This is an observation which requires further study.

The Supplementary Information (see 5, i.e. Appendix A.8 and Appendix A.9) contains similar graphs showing these calendar time data for other, younger and older, non-matched but age-stratified cohorts; and for matched cohorts as well. They all show similar patterns, suggesting similar interpretations and artifacts. We note that for the youngest group, 2000-2020, Covid-19 mortality is extremely low, and VE against ACM as calculated in this way becomes negative after approximately six months. Again, due to the suspected biases/artifacts, no clear conclusions on causality can be inferred from this.

Appendix A.11 in the Supplementary Information contains an extra analysis of the mortality patterns and calculated VE after the first *booster* round of the winter of 2021/2022. It shows similar patterns as after the primary series vaccination round. This suggests similar artifacts play a role in the ostensibly high effectiveness of booster vaccination (as reported earlier [18, 24]).

3.5. Additional statistics and findings related to the identified biases

Results. In the previous sections, we have seen various indications of both residual HVE-related confounding and vaccination status misclassification bias playing a role. In Appendix A.12 we collect and calculate various additional statistics and other information relevant to those hypotheses; both based on our own data sources and on external data sources and studies. Here we summarize the main findings.

Table 4 provides an overview of the findings, focusing in particular on what we, or other data sources and studies, estimate for older and high vulnerability individuals, in terms of their over-representation in the group with status "unvaccinated", and also in terms of the vaccination registration issues associated with those groups. The table indicates the source, whether the source supports the HVE hypothesis, whether the source shows vaccine registration problems in general, and whether the source supports the vaccination status misclassification hypothesis (i.e. asymmetric bias-inducing registration gaps). Furthermore, numerical estimates are included of the percentage of missing vaccine registrations (from different sources and calculations) and their impact on the group with status "unvaccinated". The last column shows the estimated percentage, within the status "unvaccinated" group, of people who were actually vaccinated but whose registration is missing.

Discussion. Among the calculations summarized in Table 4 and described in more detail in Appendix A.12 are calculations that indicate significantly lower (registered) vaccination rates, for high-risk individuals as compared to low-risk individuals (first row in the table). We estimated this using this study's CIMS vaccination dataset in combination with the medication-based Rx-Risk comorbidity metric. This finding is in contrast with vaccination efforts that prioritized vaccination of higher-risk individuals. Similarly, we observe (second row in the table) relatively low calculated vaccination rates in elderly persons receiving long-term care (WLZ) after vaccination round 1. This too is in contrast with what was previously reported and their priority in vaccination schedules [37]. Together with the finding that during the booster round in the fall/winter of 2021 relatively many of those same elderly appear to receive their first vaccination (see Section 3.4), this suggests relatively high numbers of missing vaccination data in vaccination round 1 for vulnerable individuals.

Table 4: Table summarizing findings and estimation w.r.t. high vulnerability individuals, their over-representation in the group with status "Unvaccinated" within age/WLZ cohorts, and their vaccination registration issues. See text for explanation.

Source	Supports HVE	Supports Vacc. Reg. Problems	Supports Misclass. Bias	Est. Ratio High Vuln. Unvacc./Vacc.	Est. % Vacc. Data Missing	Est. % Vacc. among "Unvaccinated"
CIMS, Rx-Risk (Appendix A.12)	✓	✓	✓	1.8	15 – 25%	85%
CIMS, WLZ (Appendix A.12, 3.4)	✓	✓	✓	1.6	15 – 25%	85%
OMT, RIVM ([57, 37, 58])		✓	✓		20 – 30%	
VASCO, RIVM ([31])		✓			5 – 50%	
NIVEL ([19])	✓	✓	✓	2.0 – 3.5		25 – 75%
Lareb, EMA ([59, 60, 61])		✓	✓		25 – 35%	

Combining CIMS vaccination registration data with externally sourced data on total numbers of administered vaccinations by vaccine agency as provided in public reports by Dutch national health agency and vaccination coordinator RIVM [37], we are able to estimate CIMS vaccination registration completeness by agency (see Table A.5 in Appendix A.12). We find significantly lower estimated CIMS vaccination registration completeness for vaccinations administered by agencies and parties responsible for more vulnerable groups (that is: general practitioners, hospitals, nursing homes). Specifically, we estimate 20% or more missing for those vulnerable target groups (third row in Table 4). This is much higher than the 7 to 10% missing for individuals receiving vaccinations through agency GGD. Agency GGD managed the various national vaccination centers; this is where mostly relatively healthy individuals went (typically those capable of going there using personal transport or public transit). Again, this points to substantial asymmetries in missing vaccination registrations, correlated with health. Exactly in line with this finding, early and later reports from the national Covid crisis management committee (OMT) and others in the Netherlands acknowledge and express concern about large numbers (30% or more) of CIMS vaccination registrations missing during vaccination round 1; in particular for those same vaccine administration agencies responsible for vulnerable groups [57, 37]. Vaccination registration coordinator RIVM has acknowledged that it gave up on retrieving many of these missing vaccination registrations, due to technical complexity and logistical reasons [58].

Combining these various numbers on CIMS-registered vaccinations and known or estimated missing vaccination registrations, in Appendix A.12 we calculate crude estimates of the percentage of actually vaccinated but unregistered individuals among those with status "unvaccinated" (see Table A.4, and this is summarized in Table 4, last column). Note that this percentage can quickly become large in a situation (which we have) of generally high vaccination rate, because that implies that the number of actually unvaccinated individuals is low. In our calculations (which are necessarily imprecise estimates), this percentage can be as high as 85% for older age groups; thus likely leaving only a 15% subgroup that is truly unvaccinated. Furthermore, based on similar calculations we estimate substantial higher numbers (at least 60% higher) of high risk individuals among those with status

"unvaccinated" (fourth column in Table 4).

Supportive data on this comes from various data sources and reports. Independent data from a Dutch longitudinal pre- and post-vaccination research cohort, VASCO [31] (led by RIVM), indicates significant errors and gaps in the CIMS vaccination data compared to self-reported vaccination data (fourth row in Table 4). They found that the central CIMS vaccination registry matched self-reported vaccinations completely only in about 50% of cases. Importantly, many gaps and errors were there despite (self-reported) individual data sharing consent, suggesting that consent certainly was not the only factor in the gaps and errors.

Another independent dataset, from research institute NIVEL [19] (fifth row in Table 4), contains detailed general practitioner-based care data as well as vaccination information, separate from CIMS. In that dataset, among other data they have information on the percentage of people receiving a type of care indicating very high frailty and end-of-life conditions, so-called "high-intensity care". (This is a type of data not available in our own study.) They measure and report this type of care especially in the four weeks before either an actual vaccination date (for vaccinated people), or before the expected vaccination date (for unvaccinated people). High-intensity/end-of-life care prevalence among unvaccinated people was significantly higher than among vaccinated people in their dataset: 0.2% vs. 0.1%, i.e. double the rate. For those within that small subset who subsequently died, within 12 months from their actual or expected vaccination date, differences were even more pronounced. Approx. 11.7% of unvaccinated dead individuals received this type of end-of-life care in the four weeks before their expected vaccination date, as opposed to approx. 3.4% for vaccinated individuals (i.e. an approx. 3.5 times higher rate for unvaccinated people). All this is strongly suggestive of frailty-related transient HVE that can have a high impact on short-term mortality after the first vaccination round: relatively many end-of-life individuals who no longer receive vaccinations and who die shortly afterwards.

At the same time, with respect to the vaccination status misclassification issue, that same NIVEL study confirms that many "unvaccinated" people according to CIMS were actually vaccinated (according to the GP data). Based on NIVEL's proprietary data, they were able to reduce the percentage of vaccinated but unregistered people in their study cohort from an estimated 7% to 4.5% (a relative 35% reduction). In all, they estimate that 25% or more of those labeled "unvaccinated" likely received at least one vaccination, rising to 75% in the older age groups [19] (fifth row in Table 4).

Finally, there are several studies and datasets that have focused on reported adverse events after Covid vaccination [59, 60, 61]. For the Netherlands, these data were originally collected by Dutch pharmacovigilance organization Lareb and provided to the European Medicines Agency EMA. The results (sixth row in Table 4) indicate that associated vaccination and in particular vaccine batch data, which were in most cases extracted from CIMS, was missing very often, in 25% (for serious adverse events) to more than 35% (for deaths after vaccination) of reported cases (sixth row in Table 4). Although various alternative, technical explanations exist [59], one explanation consistent with data described above is substantial CIMS vaccination status registration gaps, for these people experiencing (severe) side effects or death after vaccination.

Altogether, these additional results, based on our own data and based on external sources, amount to a substantial body of evidence supporting both the hypothesis of significant HVE-related residual confounding and the hypothesis of significant vaccine status misclassification bias. As to the latter, our estimates (see Appendix A.12), when taking various data points and data sources together, suggest that 20 to 35% of vaccinations of the most vulnerable subgroups may not have been registered, as opposed to only 5 to 10% for relatively healthy people. This could clearly lead to serious misclassification bias, due to placing relatively many very vulnerable, actually vaccinated individuals in the group with status "unvaccinated". These numbers and conclusion are notably different from the generally assumed and reported overall number of approx. 7% missing vaccination registrations (see 2.3 and [18, 30, 37, 19, 20, 21]), typically (implicitly) assumed to be uniformly and randomly distributed among different subgroups and not having a major impact.

3.6. Quantitative Bias Analysis 1

Results. The previous sections have demonstrated that both significant HVE-related residual confounding and vaccine status misclassification bias play a substantial role in explaining the results from our primary analyses. In this section, we look at the results of the first of our two complementary types of quantitative bias analysis (QBA) attempting to further understand and quantify these biases. This corresponds (see Section 2.8 and Appendix A.13) to simulation of plausible, mathematically modeled mechanisms and plausible parameters (derived from our study's data) to simulate the biases together with normal expected mortality patterns for different vulnerability subpopulations. The results are simulated KM survival data that we can compare to the measured KM survival data of previous sections, and parameters representing quantified bias size that we can inspect for plausibility.

Importantly, multiple, distinct parameter settings can produce similar simulated KM curves for unvaccinated and vaccinated groups. Each parameter setting corresponds to a different assumed situation, representing first of all a particular population composition of high- and medium-vulnerability people vs healthy people, within the vaccinated and within the unvaccinated subpopulation. The particular asymmetry in that corresponds to a healthy vaccinee effect of a certain size. Secondly, there are parameters representing particular assumptions about vaccination registration gaps. Vaccination registration gaps result (as in reality) to simulated individuals who are vaccinated ending up in the group with status "unvaccinated". When such vaccination registration gaps are asymmetrically distributed among subgroups of different vulnerability, they introduce misclassification bias. Finally, there are parameters for higher probability of death due to Covid-19 for unvaccinated people, and potentially somewhat elevated probability of death shortly after vaccination. None of these effects are 'built in' by including these parameters; the parameters are optional control variables that can be adjusted to investigate different scenarios and their effects.

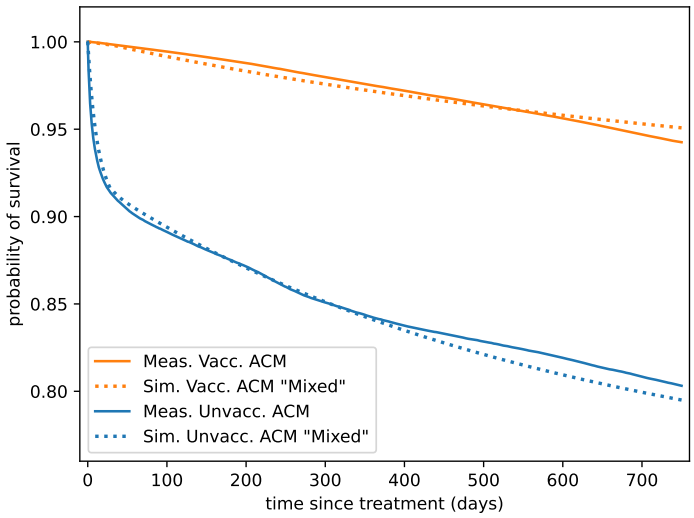


Figure 7: Measured survival curves for the 1940-1950 age cohort (no WLZ, ACM), together with *simulated* versions, as produced by the mechanistic QBA model simulation. This particular simulation is based on a mixed scenario with both a large transient healthy vaccinee effect and substantial vaccination gaps in particular for vulnerable people and especially early on in the vaccination campaign.

We focus again on the age cohort that is the main focus of this paper, the group born 1940-1950 (no WLZ long-term care). Figure 7 presents the results from one such simulation with a particular set of parameters, for this age group. Simulated Kaplan-Meier data resulting from simulated (all-cause) death rates per day, for those with (actual and apparent) status "unvaccinated" and "vaccinated", are shown together with the measured Kaplan-Meier data from the primary analysis (compare with Figure

3). This particular simulation corresponds to a "mixed scenario" with both a large transient healthy vaccinee effect and substantial vaccination gaps in particular for vulnerable people and especially early on in the vaccination campaign. Appendix A.13 in the Supplementary Information (5) presents results from other parameter settings.

Discussion. Figure 7 shows results from a "mixed scenario" simulation together with the corresponding measured results. Such a mixed scenario with both a large transient healthy vaccinee bias component and a substantial vaccination misclassification bias component seems realistic, given all primary and secondary data and analysis presented above. It leads to a good "fit" between simulated and measured results. It is also "robust", in the sense that small variations in the underlying parameters can produce similar simulation results and similar good fits, suggesting that it is not very sensitive to precise parameter settings. All this is suggestive of realism with respect to real-world circumstances at that time. This mixed scenario is characterized by a strong but transient, frailty-related healthy vaccinee bias that plays a primary driving role. This corresponds to a small but significant unvaccinated subgroup "already at death's door" driving early high mortality rates. Because it is small and death rates are very high, this subgroup depletes quickly, within a few weeks. This would explain high initial death rates in the unvaccinated followed by relatively quick attenuation of that effect. The vaccination status misclassification bias likely plays an important, related and conflated, but secondary role. The simulation results suggest that these vaccination status registration problems were probably largest during the peak of the vaccination round (the first Covid vaccination round of spring 2021), and strongly biased towards the most vulnerable groups. Without it it is difficult to realistically model the kind of extremely high mortality rate differences between those with status "vaccinated" and those with status "unvaccinated".

It is technically possible to obtain simulation results with a reasonably good fit using only limited vaccination status registration issues; but it requires an extreme HVE bias, corresponding to only very few vulnerable individuals having been vaccinated (see Appendix A.13 in the Supplementary Information, see Section 5). Such a scenario seems less realistic, given our own analyses of still many high-comorbidity individuals being vaccinated (see Section 3.5); public reports on their general prioritization and high vaccination rates [24, 37]; and independent reports based in part on separate (NIVEL) vaccination records confirming that many vulnerable and even end-of-life individuals were vaccinated according to GP records [19], despite many of those vaccination records missing in CIMS.

A completely opposite scenario of vaccination status misclassification bias being the primary driver of the simulated/observed high "unvaccinated" mortality rates, on the other hand, seems unrealistic too. Without a substantial, truly unvaccinated, high-frailty subgroup depleting at a very high rate, in our simulation model it is virtually impossible to obtain the type of simulated KM curves required; unless 90% or more of vaccine registrations for cases of death shortly after vaccination are missing, which seems unlikely.

As a sanity check, we also experimented with a simulation scenario based on some but limited HVE and negligible vaccination status misclassification bias, together with high short-term mortality in the unvaccinated (see Appendix A.13 in the Supplementary Information, see Section 5). Such high short-term mortality exclusive to the unvaccinated would presumably be due to a significant Covid wave. And it would simultaneously imply very high and immediate vaccine protection, such that vaccinated people are not affected by the Covid wave. This would be the kind of scenario compatible with reported conclusions on high Covid vaccine effectiveness following the first vaccination round in terms of all-cause mortality, implied in the earlier studies [18, 24, 19, 20, 21]. To obtain simulation results with a reasonable fit to measured results for this 1940-1950 age group, this variant of the simulation model requires parameters equivalent to a situation where a big Covid wave infects most of the people in the unvaccinated group within a period of 4 weeks or a little more. The mean Infection Fatality Rate (IFR) in that group over that period (in the spring of 2021) would have to be around

10% or more, with most deaths occurring shortly after infection. Simultaneously, it requires that most of those Covid deaths were classified as non-Covid deaths (compare with results discussed in Section 3.2 and Figure 3). Finally, vaccine protection immediately after dose 1 would have to be instantaneous and almost perfect, such that almost no one who received dose 1 ("vaccinated" in our KM analyses) are affected. All this seems unrealistic and incompatible with what we know from realistic assessments, both in our own study and elsewhere.

Various parameter settings and good-fit simulation results *are* compatible with a much smaller and more realistic vaccine protection effect, spread out over a longer-term period. This requires accepting the large biases affecting mostly the first period after vaccination. At the same time, various parameter settings and good-fit simulation results are also compatible with a small vaccine damage effect (slightly increased mortality risk after vaccination, esp. for very old and frail people), especially short-term, and possibly in interaction with a concurrent SARS-CoV2 wave driving many infections interacting with vaccinations. Such a scenario requires a substantial vaccination status misclassification effect, with relatively many deaths after vaccination being associated with a relatively large probability of missing CIMS vaccine registration. Both types of scenarios (Covid effects and positive and negative vaccine effects) are possible (also simultaneously) given our modeling and the data; but their likelihoods are unknown, given that the combined large bias effects largely mask such smaller possible effects. The mixed scenario simulation results shown in Figure 7 do in fact contain such small (both positive and negative) vaccine effects. This does not show or prove that such effects must have existed in the real world; but it does show that such effects are perfectly compatible with observed mortality patterns, and their presence would (almost) entirely be masked by the large, dominating bias effects.

3.7. Quantitative Bias Analysis 2

Results. The second Quantitative Bias Analysis largely ignores the underlying mechanisms and their plausibility investigated in the first QBA, described above. Instead, it uses a purely numerical data-driven method based on non-Covid mortality as negative control data, and numerical (proportional scaling) Kaplan-Meier data adjustment derived from that (see Section 2.8 and Appendix A.14).

Again we focus on the 1940-1950 (no WLZ) age cohort. Figure 8 shows results. The top plot visualizes the "basis" of the adjustment. This corresponds to a normalization of non-Covid mortality for either the unvaccinated group or the vaccinated group towards the (theoretically) unbiased estimate of the *overall* non-Covid mortality KM data. That overall non-Covid mortality KM survival data curve is based on a random and representative sample of the whole age group, and their combined mortality data. The correction (normalization) factors derived from this method are applied to the All-Cause mortality KM data (bottom-right plot) and to the Covid mortality KM data (bottom-left plot), to arrive at adjusted estimates.

Discussion. The top plot of Figure 8 shows that the overall non-Covid mortality KM survival curve (green) is much closer to the vaccinated survival curve (orange) than to the unvaccinated survival curve (blue). This is directly related to the fact that through the random and representative sampling, the overall non-Covid mortality data sample contains mostly (approximately 92%) vaccinated individuals. Conversely, the unvaccinated group clearly is very different from this representative overall sample from the same age group; they are the ones most affected by the biases. The Kaplan-Meier data of the (small) unvaccinated group is affected disproportionately by the systematic biases that are, in turn, most likely caused by a small but significant and influential subset of high death rate individuals. Accordingly, the necessary bias adjustments are much larger for the unvaccinated group than for the vaccinated group. These are shown in the bottom plots of Figure 8. These adjustments correspond to time-varying proportional adjustments of the original KM data.

Importantly, the adjustment for the all-cause KM data does not simply 'remove all non-Covid mortality data' from the all-cause mortality data. (Doing so would simply leave only Covid mortality

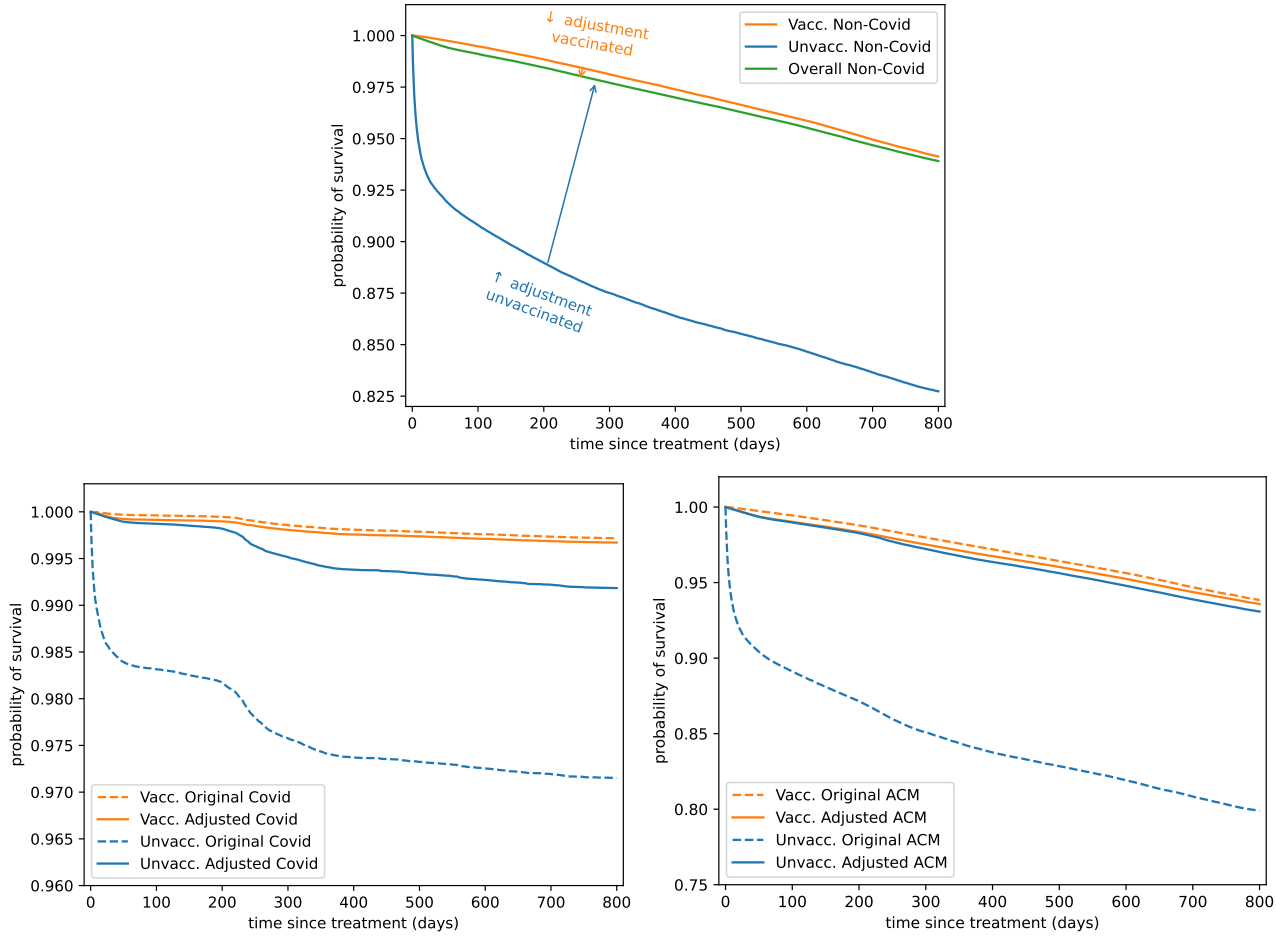


Figure 8: Original and QBA-adjusted survival curves of vaccinated and unvaccinated individuals of age cohort 1940-1950, no WLZ use. The adjustments are based on using non-Covid mortality data over time as the negative control data, and the adjustment to the unbiased overall estimate is shown schematically in the top figure. The KM survival data adjustment for mortality from Covid is shown in the bottom-left figure; the KM survival data adjustment for all-cause mortality (ACM) is shown in the bottom-right figure. Note the different scales on the y-axis.

data in the all-cause mortality data.) Instead, it removes the *bias* observed in the non-Covid mortality data; by looking at the likely bias-induced and strongly time-variant *differences* in non-Covid mortality between vaccinated and unvaccinated groups. Again, as explained in Section 2.8 and Appendix A.14, this bias adjustment does not claim perfect accuracy or validity. It is based on a particular bias adjustment model, built on assumptions. It makes precise what happens in terms of adjusted KM survival data (i) if one accepts non-Covid mortality as a good and reasonable negative control; and (ii) if one accepts a straightforward multiplicative (proportional) correction factor based on that. In summary, for both all-cause and Covid mortality, we expect the bias adjustment to be a reasonable but only approximate correction, in line with similar methods from the literature [33, 50, 51, 52, 53].

For the ACM KM data (bottom-right plot), we observe that the adjustment removes almost all of the apparent protective effect of the vaccine, mainly by removing most of the unrealistic (biased) high unvaccinated mortality. The adjustment is largest early on (first period after start of follow-up), in line with the earlier explanations of the biases and corresponding data, and in line with the QBA simulations of the previous section. In absolute terms (ARR), a protective effect of approx. 0.5 percentage point in all-cause mortality survival remains for the vaccinated group, after 800 days (compared to 14 percentage points in the original KM data). This corresponds to relative effectiveness

against death (VE_{RRR}) of 7% (down from 70%). No advantage is observed for the first 250 days, i.e. approximately the whole period from vaccination in spring 2021 until the very end (and winter) of 2021. The adjusted KM data for Covid mortality (bottom-left plot of Figure 8, note the different scale) mirror these results. Similar to the ACM results, approximately 0.5% in terms of absolute Covid mortality protection remains for the vaccinated group, after adjustment, after 800 days (down from 2.6 percentage points). And similarly, adjusted protective effects emerge only after 250 days, i.e. around November/December 2021 (when a new Covid wave came, corresponding to the tail of Delta and the arrival of Omicron).

Thus, if one accepts the bias adjustment model, these combined adjustment results suggest limited but still some actual vaccine protection; but much smaller than has previously been claimed. This seems realistic; but at the same time, these results are uncertain, and we must (once more) acknowledge limitations of the heuristic adjustment method, in either direction. The method depends on non-Covid mortality being clean and independent and suitable for this adjustment approach. More subtle healthy vaccinee or selection effects not captured by clear differences in non-Covid mortality cannot be resolved by this bias adjustment method, by definition. This could for example include immunocompromised patients who, for that reason, could not be vaccinated; and who are also extra susceptible to SARS-CoV2 infection and subsequent death. Also, differential booster (typically third dose) vaccination uptake by different groups may lead to an additional selection and healthy vaccinee effect, which cannot be captured by this relatively simplistic bias adjustment model. Most importantly, both SARS-CoV2 infection's acute or non-acute impact on non-Covid mortality *and* potentially any vaccine-associated negative impact on non-Covid mortality become invisible, using this bias adjustment method. Thus, if non-Covid mortality data contains relatively many deaths that were actually from Covid but that were misattributed [40, 62], then the bias adjustment from our method would be too large. This would lead to underestimation of the "true" vaccine effectiveness profile. If, alternatively, the non-Covid mortality data contains relatively many deaths which can and should realistically be attributed (at least partially) to the Covid vaccines [24, 59, 61, 63], then the bias adjustment from our method would be too small. This would lead to overestimation of the true vaccine effectiveness profile and safety profile.

For younger age cohorts, calculated bias adjustments are qualitatively similar; but the remaining calculated vaccine protection becomes ever smaller. This is in line with previously discussed observations on Covid mortality becoming an ever smaller factor for younger cohorts, both in absolute terms (absolute risk of death from Covid), and relative to other causes of death. With bias according to non-Covid mortality differences remaining large, bias adjustment for these younger cohorts results in ever more similar vaccinated and unvaccinated KM curves. For the 1950-1960 cohort, after bias adjustment 0.1 to 0.2 percentage points of apparent survival advantage remains for vaccinated individuals, after 800 days. For even younger cohorts, the adjusted apparent vaccine protection effects become virtually non-existent. For the oldest and most frail cohorts, substantial apparent vaccine protection after this bias adjustment does remain; see results in Appendix A.14. For the 1940-1950 cohort *with* WLZ (long-term care, typically nursing homes), apparent absolute vaccine protection against death after adjustment, after 800 days, is around 5% (down from 27%); with relative protection, VE_{RRR}, of around 11% (down from 40%). However, by that time, almost half of all individuals in the group, either vaccinated or unvaccinated, have died. Thus, the selection effects described above, which presumably limit the applicability of the adjustment method, can clearly play an important role here. Moreover, even more than in the healthier no-WLZ group, we see a clear signal of temporarily increased overall non-Covid mortality for the WLZ group, exactly during and right after the vaccination round. This could either be caused by undiagnosed Covid deaths in the non-Covid category, or non-Covid deaths with a (partial) causal link to the Covid vaccines (as in Norway [63]); either of which partially invalidates this particular bias adjustment method.

4. General discussion

4.1. Main findings

In the wake of the first waves of the SARS-CoV-2 pandemic and thereafter, many scientists were surprised that mortality in many Western countries, including The Netherlands, increased or remained elevated from 2021 and onward, despite the large-scale roll-out of vaccines [64, 65, 62, 66, 67, 68, 69, 70]. In some countries such as Australia, New Zealand, Japan, Hong Kong and Singapore, which mostly avoided large SARS-CoV-2 infection waves until relatively late, and after most of the population was vaccinated, excess mortality even only started at that late stage [71, 72, 73, 74, 75, 76]. One would have expected a steep drop in excess mortality since vaccine rollout, and/or little or no excess mortality in those countries like Australia and New Zealand, but for the most part this expectation did not materialize. In those studies, various reasonable hypotheses on likely reasons are discussed, and countries like Australia and New Zealand do appear to have relatively low cumulative excess mortality; but in any case, this has renewed interest in appropriate assessment of the extent of the protective effects of Covid-19 vaccination on all-cause mortality. However, observational vaccine studies, that we must rely on for this, are plagued by confounding and biases. In this study we have re-analyzed Dutch observational data that was previously reported to show very high VE, even in terms of non-Covid and all-cause mortality [18, 19, 20, 21]. As primary methods we used standard analysis methods, including stratification and matching-based survival analysis. Furthermore, we used both standard and non-standard methods and analyses to check and correct for various well-known types of biases.

We find that very high apparent VE, as calculated directly from relative mortality rates, results primarily from extremely and unrealistically high mortality among unvaccinated individuals in the first weeks since the start date of follow-up (corresponding to dates during the first vaccination round), gradually decreasing afterwards. This is accompanied by simultaneous decreased mortality (compared to normal baseline rates) among vaccinated individuals, in the first weeks after vaccination.

These findings cannot realistically be attributed to direct vaccine efficacy against Covid-19, as the extreme differences start immediately after dose 1, and remain when considering only non-Covid-19 deaths (the latter corresponds to a negative control sanity check [54]). Given widespread testing during these periods, it is unlikely that such high numbers of Covid-19 deaths would have been missed or been misattributed (see Appendix A.3). By far the most frequent primary recorded cause of death for the unvaccinated subgroup in that first vaccination round is cancer (see Section 3.4), and its relative rate was higher among those deaths, when compared to baseline causes of death patterns. This also argues against Covid-19 as the principal driver of these elevated mortality rates. Furthermore, as is apparent from the calendar time analyses, these elevated mortality rates in unvaccinated groups are observed mostly in the spring and summer, seasons which are not normally associated with high mortality from respiratory viruses (another variation of a negative control sanity check).

It is similarly unlikely that the differences result from vaccine efficacy against medium or long-term complications (post-acute sequelae) of Covid-19. Such effects would most likely be seen starting several weeks after vaccination, while our data show extreme differences shortly after vaccination (dose 1) that largely disappear after a few weeks. It is theoretically possible that the vaccines have an unexpected short-term effect that greatly reduces mortality independently of their effect on Covid-19. Indeed a broader effect of these vaccines on the immune system has been described previously [77, 78]. However, to date there is no plausible biological mechanism that could cause such extreme effects at such short time scales. And most importantly, this hypothesis does not explain why mortality in the *unvaccinated* group increases compared to normal values, while this group receives no intervention. We conclude that the large differences in mortality between unvaccinated and vaccinated individuals most likely result primarily from residual confounding and/or other systematic biases.

4.2. Residual bias: healthy vaccinee bias

Our methodology should have adequately corrected for immortal time bias. Our matching strategy also appears reasonable and effective (see Section 3.3 and Appendix A.5); but residual healthy vaccinee bias seems very probable. This healthy vaccinee bias constitutes in particular *time-varying, short-term* HVE (see [25, 27, 29, 79]), almost certainly related to people who are not vaccinated due to already being terminally ill or hospitalized at the start of the vaccination round (sometimes also referred to as "high death risk group", or a "frailty bias"). Perhaps unsurprisingly, that information is not necessarily captured well by medication use in the previous year, which we used for comorbidity-based matching; and our dataset does not contain information to accurately identify that subgroup. A relatively large percentage of all deaths likely comes from that small subgroup, affecting all results accordingly.

Evidence and reports from secondary, independent data sources supports this hypothesis (see Section 3.5). While this HVE artifact is strongest in the first weeks since vaccination, it likely persists longer-term and gradually dissipates in a way that is difficult to estimate. Next to impacting the KM data and derived VE metrics for non-Covid and all cause mortality, this HVE bias likely affects the Covid mortality and Covid VE calculations as well. This is likely the case because Covid mortality is strongly related to underlying comorbidities affecting all mortality, and the Covid and non-Covid KM curves all show similar patterns and appear to be strongly correlated (patterns corroborated by the calendar time analyses).

This large residual short-term, HVE-related confounding effect suggests that we are dealing with what is actually a large confounding-by-indication effect [27, 29, 80, 79]: the fact *that* someone has not been vaccinated may in itself be an important indicator or predictor of potentially very high vulnerability, even when other, available individual-level data provide no such information on underlying high vulnerability. This represents a selection bias for which, we believe, it is virtually impossible to correct given imperfect and incomplete data for individuals (Appendix A.10 contains a limited additional analysis of this issue, including calculations to understand and quantify part of the confounding-by-indication effect).

4.3. Residual bias: vaccination status misclassification bias

Besides residual HVE, the extreme size of the short-term mortality effect for the unvaccinated group (despite matching), as well as secondary and supportive analyses described in Section 3.5 and later, strongly suggest that there is also another factor at play: vaccine status misclassification bias. The national registry of vaccination status CIMS is incomplete and the group labeled as "unvaccinated" actually consists of both unvaccinated persons and many vaccinated persons who did not give consent or were otherwise not included in the database. As described in Section 2.3, Section 3.5, Appendix A.2, and Appendix A.12, missing or inaccurate vaccination data clearly does not result from lack of individual consent alone, and in any case is unevenly distributed over different subgroups. Omission from the national vaccination registry CIMS almost certainly occurred much more frequently for frail individuals and/or individuals who died shortly after vaccination, and this has likely contributed significantly to the systematic bias effects.

Methodologically speaking, this type of data integrity problem and associated misclassification (or information) bias goes beyond "classical" residual/unmeasured confounding, which is typically construed and interpreted as caused by imperfect predictive information captured by covariate data, and results in some (typically limited) unexplained (residual) variance. Instead, this type of misclassification bias is, in a way, more fundamental. It affects group assignment itself, and leads to potentially highly asymmetric and nonlinear unexplained variance that completely invalidates classical methods used elsewhere [18, 24, 19] such as regression methods, or other methods that assume relatively unbiased group assignment and high data integrity; and it is typically not easily identifiable using those standard methods. Furthermore, we observe that the specific form of vaccination status misclassification bias identified in this paper goes further than the other, similar misclassification bias forms

highlighted in some previous studies of Covid vaccine effectiveness [32, 29, 81, 33]. Those studies studying or mentioning similar misclassification issues mostly focus on the potential bias introduced by not considering individuals as "fully vaccinated" until two weeks after completion of the full (typically two-dose) primary Covid vaccination course. Our analyses strongly suggest even more basic data integrity issues: in our dataset, even dose 1 (and any dose) vaccination registrations are missing in substantial, asymmetric and strongly bias-inducing ways.

Our main results as well as secondary evidence are highly consistent with, and suggestive of, this data integrity issue and associated misclassification bias hypothesis. Similar to the HVE artifact, this misclassification artifact likely has the strongest impact in the first weeks since vaccination, but it persists longer-term in a way that is difficult to estimate. In general, the impact of vaccine status misclassification on calculated VE may go either way, depending on the composition of those vaccinated but unregistered people. The authors of the NIVEL study described earlier [19] as well as two RIVM studies [18, 30] suggest an attenuating effect, leading to underestimation of true VE, due to the vaccinated but unregistered subgroup potentially contributing generally healthier (vaccinated but privacy-conscious) people to the "unvaccinated" control group. This effect seems plausible, in principle. However, both the primary results from our study and the secondary data described above and in Appendix A.12 suggest a much stronger opposite effect. The evidence shows first of all that absence of registration is often unrelated to explicit consent. And almost certainly a relatively large proportion of the vaccinated but unregistered subgroup are from very vulnerable groups; those contributing most to the high death rates observed in the group with status "unvaccinated".

4.4. Conflated bias effects and the general issue of large heterogeneity

Importantly, the two major artifacts we have identified are not independent from each other. Instead and as pointed out above, they are likely related and conflated; and this is probably why their impact on all results is so large. Those individuals who were most vulnerable, and who therefore contribute most to the observed residual short-term HVE effects, first of all had a relatively high probability of not being vaccinated due to being too weak. And secondly, if nevertheless vulnerable people *were* vaccinated, all evidence suggests that they had a relatively high probability of their vaccination not being recorded in the central vaccination database CIMS. The latter category perhaps included some people who died shortly after vaccination, due to patient records being closed or related reasons, as has been reported in the UK [32]. The combined effects lead to many vulnerable people, and many deaths, ending up in the "unvaccinated" group.

In any case and regardless of precise numbers and effects, we can conclude that heterogeneity within the cohort that has status "unvaccinated" is very high. Among them, a relatively high proportion was unvaccinated due to health reasons (already in end of life conditions and the like). And also among them, a relatively high proportion was in fact vaccinated but their vaccination records are missing, and they also disproportionately come from vulnerable groups. The proportion of 'normal, comparable, relatively healthy' people who made a conscious choice to not get the vaccine (and who would normally be the control group that one is most interested in as a fair comparison), is low: in our estimates as low as 15 to 25%, in older age cohorts.

All of this makes drawing conclusions on true causal effects of the Covid vaccines, from our observational study and others that depend on the same or a similar dataset, very difficult.

4.5. Masking of potential safety signals

The previous Dutch studies performed on the same dataset [18, 19, 20, 21] explicitly included and looked at non-Covid mortality effects shortly after vaccination, in part because of some and mostly anecdotal concerns about Covid vaccine safety [24, 63]. Those studies unanimously conclude that such concerns are unwarranted, due to non-Covid mortality shortly after vaccination being significantly lower than normal, baseline mortality, and much lower than mortality for unvaccinated people in

the same period. However, in line with the conclusions described above for vaccine *effectiveness*, our results and conclusions on identified large biases apply equally to vaccine *safety*: short-term non-Covid mortality patterns for both vaccinated and unvaccinated people shortly after vaccination are masked by these large artifacts. It is possible that a smaller safety signal is masked by a larger bias effect in the opposite direction.

In fact, some preliminary, exploratory observations (described in Sections 3.4 and 3.7) raise questions: for some age groups, especially old and vulnerable ones, combined *overall* non-Covid and all-cause mortality rates are somewhat elevated during and immediately after the vaccination rounds of 2021, compared to normal baseline mortality rates (e.g. see Figures 5, 6, A.10, A.11, row B; and Figures 8 and A.26). This pattern is visible for dose 1 and 2 (spring of early 2021), and also the first booster, dose 3 (autumn or winter of 2021/2022), whenever the particular vulnerable age group is vaccinated. This pattern is typically not in line with the usual seasonality (winter) mortality pattern, and shifts according to when a particular age group had its vaccination peak. The increase in deaths is often seen in both the category of unvaccinated and the category of vaccinated people. It also coincides with somewhat increased Covid-19 mortality in both groups. In other words, there is a temporal relationship between vaccinations administered in a given period and increased Covid-19 and other mortality, which cannot easily be explained by the observed biases, the winter season, Covid-19 alone, or other factors.

This does not necessarily correspond to evidence for large-scale vaccine-related deaths, and conceivably the slightly elevated mortality rates can all be explained by increased concurrent SARS-CoV-2 prevalence, in line with the observed increase in Covid-19 deaths. However, the results are suggestive first of all of poorly understood relationships between vaccination rounds and increased prevalence of SARS-CoV-2; poorly understood risks of planning a vaccination round during high SARS-CoV-2 prevalence; and poorly understood interactions between, and potential risks associated with, a person's vaccination moment and their very recent, or subsequent, SARS-CoV-2 infection. Such relationships and interactions are known to exist [82, 83, 84]. Further study may thus lead to better understanding of the optimal timing of vaccination rounds, better selection of vulnerable groups to be vaccinated and when to vaccinate them (or not), or more strict exclusion criteria or delay of vaccination depending on, e.g., recent or suspected viral infection.

4.6. Insights from our quantitative bias analyses

We used two complementary quantitative bias analysis (QBA) methods to better understand, model, and quantify the biases. The first method modeled the biases in terms of likely mechanisms and parameters (denoting their size and duration etc.). The results from these simulations suggest (or rather: confirm) first of all that Kaplan-Meier survival curves such as measured in our study cannot realistically be obtained without the presence of large biases of the types discussed above. They can almost certainly not be the result of Covid-related and/or vaccine-preventable deaths in the unvaccinated groups (such death rates would have to be extremely high and somehow peak exactly during a vaccination round). In other words, using this model, no realistic simulated mechanisms and parameters can produce results consistent with strong vaccine protection only, in line with the data we observe in the primary KM analysis.

Secondly, these simulation results suggest that the hypothesized frailty-related healthy vaccinee bias likely played a primary role, corresponding to a small but significant and quickly depleting unvaccinated subgroup "already at death's door" driving early high mortality rates. The vaccination status misclassification bias likely played an important, related, secondary role. Without it it is difficult to realistically model the kind of extremely high mortality rate differences between those with status "vaccinated" and those with status "unvaccinated". All this is consistent with what we know about (i) the significant effort and prioritization of vaccination of high-vulnerability individuals during the first vaccination round; (ii) the likely exclusion of some but not all of the most frail individuals during

vaccination; (iii) the significant vaccination registration problems for high-vulnerability individuals during the first vaccination round.

Finally, these mechanistic QBA simulation results can also be reconciled (in terms of possible and plausible parameters) with both a real but relatively small protective effect of vaccination against Covid death *and* with a potential, real but small increased mortality risk after vaccination for vulnerable subgroups; both of which would be largely masked by the large bias effects.

The second QBA method mostly ignored underlying mechanisms and was data-driven, producing heuristic but reasonable, empirically grounded bias adjustments. It used non-Covid mortality as negative control data and calculated and applied correction factors based on that. Such hypothesized adjustments, when accepted and applied, lead to substantially lower and even close to zero estimated vaccine protection against death, and (thereby) an uncertain risk-benefit profile for subgroups. This bias adjustment does not claim perfect accuracy, but it makes precise what happens in terms of adjusted KM survival data if one accepts non-Covid mortality as a good and reasonable negative control, and if one accepts a straightforward multiplicative (proportional) correction factor based on that. In other words, it calculates and makes precise what happens if one follows the intuition that apparent non-Covid mortality protection from the vaccines must largely be bias-driven. In any case, we believe that these bias-adjusted results, while imperfect, are probably closer to the "true" vaccine effectiveness profile (whatever that is) than what is obtained from both our own direct primary analyses and those of previous studies based on the same dataset [18, 24, 19, 20, 21], which largely ignored, underestimated, or minimized these large systematic biases.

One can reasonably argue (and studies have argued [40, 65, 67, 68, 69]) that causes of death, and their records, are not as independent, clean and clear as this negative control data approach assumes. This would mean that perhaps some non-Covid mortality actually was at least partially related to both acute and post-acute SARS-CoV2 infection effects, and that non-Covid mortality cannot be used as a clean negative control. However, at the same time this would mean that focusing on Covid mortality as the single primary endpoint in studies of this kind would be insufficient. And this would imply acceptance of one core argument and starting point of our paper, which is that inclusion of all-cause mortality (and/or morbidity) endpoints and metrics is necessary in most studies; and that addressing unusual ACM effects is crucial to full understanding of study results.

In any case, we acknowledge this important and inherent limitation of this data-driven bias adjustment method based on non-Covid mortality as negative control data. In fact, both SARS-CoV2 infection impact on non-Covid mortality *and* potentially any vaccine-associated negative impact on non-Covid mortality become invisible using this bias adjustment method. Therefore, its results must be interpreted with caution, similar to most of the other results from this paper.

4.7. Strengths and limitations

The key strength of our study is that our approach, based on standard methods, matching, and extensive sanity checks, visualizations, and secondary analyses, is very complete and transparent about

1. what measured mortality and survival patterns over time look like, for carefully matched vaccinated and unvaccinated groups;
2. realism of measured mortality patterns when compared to normal baseline mortality patterns, expected seasonality effects, and expected causes of death patterns;
3. the likely existence of large systematic biases and their likely causes and approximate sizes;
4. the gaps in our data and in the knowledge that we can extract from that data.

The most important limitation of our study, and at the same time the most important conclusion, is that despite our broad methodology and the multiple ways of analyzing the data, the anomalies discovered in the dataset prevent us from drawing definitive conclusions about the effect of Covid-19 vaccination on all-cause mortality. In our view, the large systematic residual biases, and especially the

significant and likely biased vaccination registration data gaps, make it virtually impossible to accurately correct for the biases and estimate reliable, accurate VE estimates, given the current dataset. A related limitation is that these large systematic biases make it impossible to use standard epidemiological statistical significance tests or Bayesian likelihood estimates.

4.8. Main conclusions

Realistic Covid-19 vaccine effectiveness and safety profiles for different age groups, after the first Covid vaccination round in 2021, are masked by the two large artifacts of HVE and misclassification. A smaller but similar analysis of the first booster vaccination round at the end of 2021 (see Appendix A.11) suggests that the same artifacts artificially inflate apparent VE [18] during that booster round, but probably to a lesser extent.

When taking into account the detected artifacts, which lead to overestimation of VE, VE with respect to ACM appears at best to be considerably lower than previously described, or even close to zero. In other words, most of the protective effect from the vaccines disappears if one tries to adjust for it (in a reasonable way, using quantitative bias analysis, see Sections 3.7 and 4.6). The (imprecise) adjusted results suggest that there may still be a limited positive but uncertain protective effect against Covid death, mostly due to the winter of 2021/2022, for vaccinated groups from older age cohorts. However, for younger age groups (roughly those born 1960 or later), estimated vaccine protection against death is (even) more uncertain and very close to zero, leading to an uncertain and undecided vaccination risk-benefit profile. On the other end of the spectrum, for the very oldest, most frail subgroups, an apparent protective effect against Covid death exists after adjustment; but data on their frailty status and data on their vaccination status is least complete and reliable, and the analyses are most uncertain. Also, small but significant safety signals exist for those most vulnerable subgroups, such that for them too the vaccination risk-benefit profile is uncertain and should perhaps always be weighed carefully on an individual basis. Various analyses in this paper, including the QBA bias calculations and adjustments, provide support for these conclusions; but in any case they remain inconclusive.

In line with these conclusions, our findings also shed light on the discussions around the reasons for persistent excess mortality in many Western countries even after large-scale Covid vaccine rollout, referred to above. Such persistent excess mortality after and despite high vaccination rates becomes less surprising if one accepts our conclusion that in any case real-world vaccine effectiveness with respect to ACM was at best considerably lower than previously expected and described.

We find no evidence for large-scale vaccine-related deaths (which has been claimed elsewhere [85, 76]). But there are small safety signals, and limited harm could largely be masked by the large observed biases in the opposite direction (see Section 4.5). Equivalently and conversely, genuine substantial benefits may be present but they have largely become invisible and uncertain due to the large artifacts.

Despite being unable to draw definitive conclusions on Covid vaccine effectiveness and safety, our findings warrant reporting as in any case they contradict previous reports about this same study population and dataset. Those reports, produced by Dutch public health authorities and other research groups [18, 24, 38, 19, 20, 21], overwhelmingly concluded very high vaccine effectiveness and safety, not only in terms of Covid-19 mortality but in terms of ACM as well. Our results suggest that their analyses were affected by the same large systematic biases – biases that, although acknowledged to some extent, were not appreciated fully in terms of their huge impact on all results. We cannot directly compare numerical results with those previous studies, due to differences in methodology and statistical analysis. Our somewhat unusual methodological design choice of follow-up immediately after dose 1 (instead of two weeks after completion of the primary course) is, however, mirrored in (typically secondary) analyses by several other Dutch studies [18, 24, 38, 20], and similar results are found throughout. Taken together and given the similarity of numerical results found by all

studies, we conclude that almost certainly the unusual results are not strongly dependent on (our or others') specific methodological design choices, but are due to the biases that are intrinsic to the study population and dataset, which was used by all.

Studies in other countries and on other datasets may suffer from similar biases and artifacts. This seems likely, in fact, given the widespread observation of extreme beneficial effects of Covid-19 vaccination on non-Covid and All Cause mortality [22, 23]. And similar biases and artifacts, especially HVE, have indeed been suspected and reported in multiple studies [23, 25, 32, 81, 26, 53, 33, 79].

Our results included and compared Covid, non-Covid, and all-cause mortality data, and both relative and absolute risk reduction metrics (RRR and ARR, respectively). Irrespective of calculated VE estimates (which we know to be imprecise), these metrics demonstrate that in relatively younger cohorts, roughly speaking people born 1960 or later, Covid-19 mortality in the Netherlands (according to official causes of death data) comprised only a very small proportion of total deaths. This holds true for both vaccinated and unvaccinated individuals. The low absolute risks of Covid-19 and low absolute risk reduction, and thus low potential benefit of vaccination in terms of death risk, in combination with what we know about risk from side effects [59, 8, 9, 63], underscore the importance of broad risk-benefit studies based on robust data. These are key to discussions about public health policies related to vaccines for those younger groups.

4.9. Future work: planned follow-ups based on the same and related data

The results warrant further study, in part to see which of the identified biases plays the largest role, and to see whether some of these biases may be resolved or reduced, so as to obtain better Covid-19 vaccine effectiveness and safety results. Plans for follow-up work are underway.

Given the identified limitations outlined above, we believe it is impossible to accurately correct for the identified residual biases/artifacts and recover reliable VE estimates using this exact same dataset (despite the otherwise high quality of the Dutch nationwide CBS microdata databases [35, 36]). Even our quantitative bias analyses, although relatively sophisticated and instructive, have clear limitations; the biases cannot accurately be resolved. Thus, one important follow-up step in our planned future work is extension of the dataset with additional, fine-grained data that may shed light on and reduce these two identified major biases. I.e., this focuses on including additional data on diagnoses or types of care suggestive of end of life and/or high vulnerability conditions; as well as additional, independent, more complete vaccination data (perhaps for a subset of Dutch individuals).

Furthermore, there are other, secondary open questions based on our observations that warrant further study and that we plan to investigate in future work. These include the preliminary observation of elevated mortality for some (especially very old) subgroups during vaccination rounds, coinciding with somewhat elevated Covid mortality, described in 4.5; this may point at poorly understood interactions between recent vaccination and infection or vice versa, and potentially an associated interaction or vaccine risk. Another avenue of follow-up research is survival analysis using other endpoints than mortality data, i.e. disease diagnoses and/or morbidity metrics, to have broader assessments of long-term vaccine protection and other effects.

4.10. Future work: general lessons

In the past decade, multiple studies have observed that vaccines effective against a single disease may not have positive effects on all-cause mortality, or vice-versa, have larger than expected positive effects. Several mechanisms could explain these counter-intuitive observations, such as: (i) direct vaccine adverse effects may equal or outweigh positive benefits, (ii) the vaccine may offer protection against a single pathogen strain but a different pathogen may simply take its place, or (iii) vaccines can have non-specific immunological effects (both positive and negative). Regardless of the underlying mechanism, these findings have challenged the standard paradigm and suggest there is a pressing need for better vaccine studies that take this into account [6, 10, 7]. In particular, studies should go beyond

singular target disease endpoints and more comprehensively take into account all-cause disease and mortality.

Moreover, with respect to the Covid-19 vaccines (but also some other vaccines), we rely primarily on retrospective observational studies to inform us on real-world effectiveness and safety. However, observational vaccine studies are known to suffer from healthy vaccinee bias, misclassification bias, and other biases [23, 27, 28, 29, 54, 81], and we find the same issues here. These are in fact issues that have been recognized for decades [86]. Future epidemiological work should put renewed focus on how to best address this. In any case, our results highlight the need for sufficiently fine-grained individual-level data on comorbidity and vulnerability that go beyond typical basic electronic health records data. They also highlight the need for high data integrity of data on exposure or treatment (in our case, vaccination) and outcomes, and careful checks focused on data integrity. If (and only if) those conditions are met, retrospective observational epidemiological studies can work as cornerstones of public health policy evaluation.

As for analytical approaches, studies should include the more frequent and stringent use of negative controls analysis [54], and other sanity checks, including stricter checks of the proportional hazards assumption and the assumption of homogeneity of research groups and treatment effects [56]. Furthermore, potentially sophisticated methods and frameworks following target trial emulation principles [87, 88], (other) causal inference approaches [43], and sophisticated quantitative bias analysis methods [49], can be employed. All these methods and approaches are intended to assess and improve plausibility of calculated results, to detect and where possible quantify or prevent residual confounding and systematic biases, and in general to improve the validity of inferred causal effects of treatments or exposures. If these biases cannot be adequately corrected for, as may be the case in certain situations, then other avenues should be explored, such as increasing the size, scope and duration of phase 3 clinical trials, or using additional (imperfect yet useful) real-world cluster-randomized implementation trials after clinical trials [13], to provide additional assessments and safeguards.

5. Supplementary information

This article has accompanying Supplementary Information, Appendix A, available online.

6. Funding

Bram Bakker and Jona Walk received funding from a Dutch crowd-funding project on excess mortality and vaccination organized by the "De Menselijke Maat" Foundation and Ronald Meester and Marc Jacobs, as described in https://stichtingdemenselijkemaat.nl/academisch-onderzoek-naar-een-mogelijke-relatie-tussen-vaccinatie-en-oversterfte/.

Bram Bakker also received funding from ZonMw (Care Research Netherlands in the Medical Sciences), through project 10430252220015, led by Eline van den Broek-Altenburg, whose support and advice are gratefully acknowledged. This was a project within the ZonMw "Covid-19 and Excess Mortality" research program (https://www.zonmw.nl/en/research-reasons-and-causes-excess-mortality-during-coronavirus-crisis).

7. Ethics approval

This is an observational study. Under Dutch law and through approval from both the Dutch ZonMw national Excess Mortality research program (https://www.zonmw.nl/en/research-reasons-and-causes-excess-mortality-during-coronavirus-crisis), and approval from the CBS Microdata Research Environment [34], no further ethical approval was required.

8. Author contributions

Bram Bakker and Ronald Meester conceived the study. Bram Bakker performed the analysis with input from Jona Walk and Ronald Meester. The first drafts of the manuscript were primarily written by Bram Bakker and Jona Walk and critically reviewed by Ronald Meester, later iterations were adapted incrementally by all authors. All authors have read and approved the final manuscript.

9. Data and code availability

This research was performed in the CBS Microdata Research Environment, which does not allow full exports and open sharing of all source data at the level of individual citizens, anonymized or not, due to privacy-related laws and regulations, in particular EU GDPR. Researchers wishing to access these same data may do so by applying for a project at the CBS Microdata Research Environment [34], as explained here: https://www.cbs.nl/en-gb/our-services/customised-services-microdata/microdata-conducting-your-own-research.

However, all aggregated data exported by us in accordance with the CBS Microdata data export rules, as used and (partially) shown in this paper, are available for researchers. Similarly, all code and scripts used in the analyses described in this paper were exported from the CBS Microdata environment, and are available for researchers. The same holds for the scripts used for post-processing and quantitative bias analysis. All data and code and scripts are available through this public repository: https://github.com/brb1901/Research-biases-in-Dutch-observational-health-data-in-relation-to-Covid-19-vaccines.

Almost all preprocessing and analysis scripts were written in Python, using a combination of custom developed code and standard libraries, including in particular the Pandas library (https://pandas.pydata.org/), the Lifelines survival analysis library https://pypi.org/project/lifelines/, and using the Anaconda data science and AI framework (https://www.anaconda.com/). Rx-Risk Comorbidity Index calculation was done using our own implementation based on the open source R scripts at https://github.com/ropensci/coder. The calculations and quantitative bias analyses of Sections 2.8, Appendix A.10, Appendix A.12, Appendix A.13 and Appendix A.14 were performed using a combination of custom Python and Excel scripts.

10. Declaration of Competing Interests

Jona Walk has conducted research on vitamin K-dependent proteins in Covid-19, which was funded in part by a grant from the Canisius Wilhelmina Hospital, and discloses application of a patent on vitamin K related to potential Covid-19 prevention or treatment held by KOVIT BV, of which Jona Walk is part owner. Bram Bakker and Ronald Meester declare no competing interests.

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